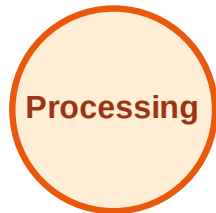


BFS Traversal

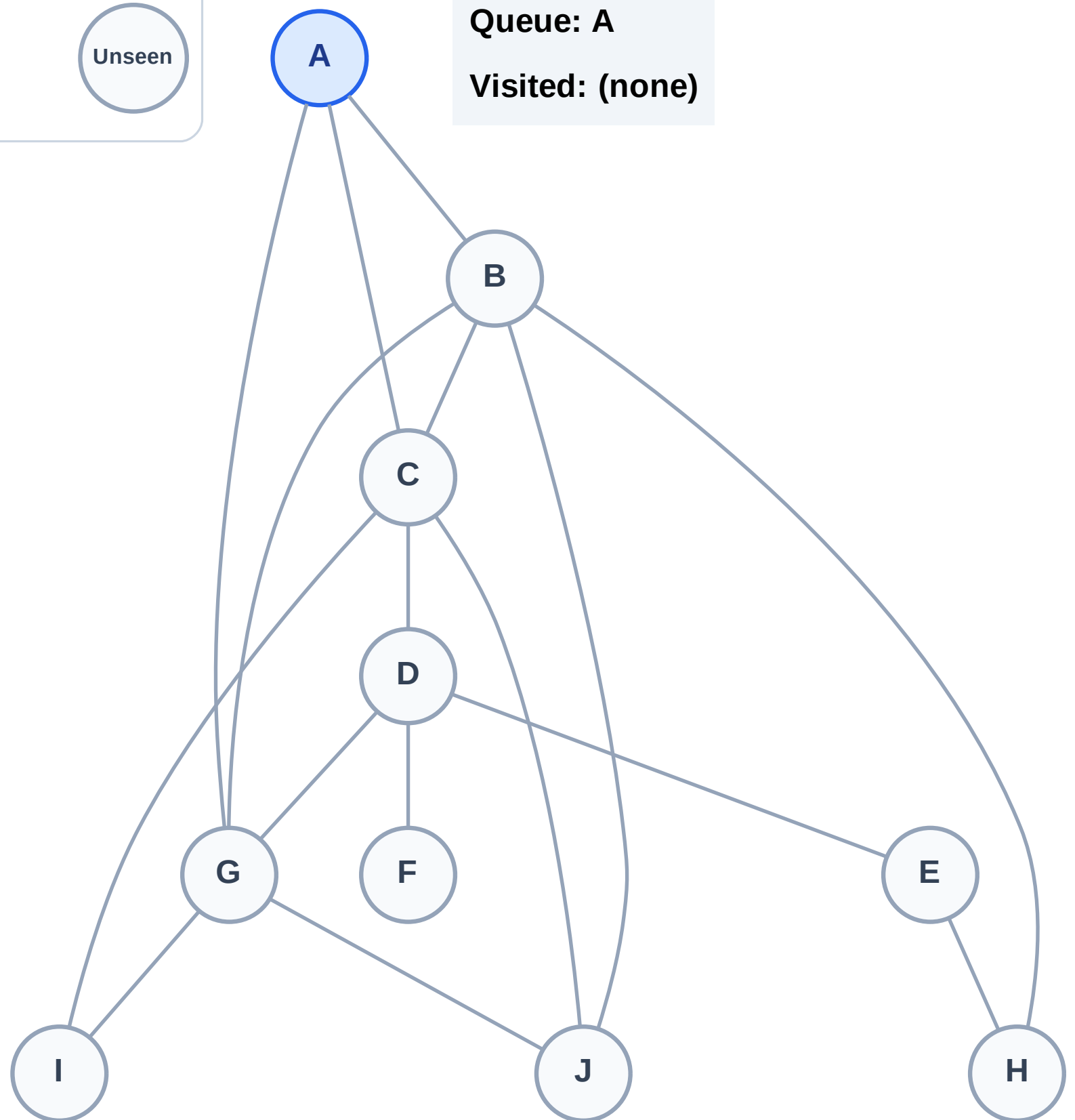
Step 1: Start at A

Legend



Queue: A

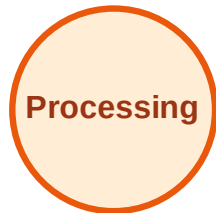
Visited: (none)



BFS Traversal

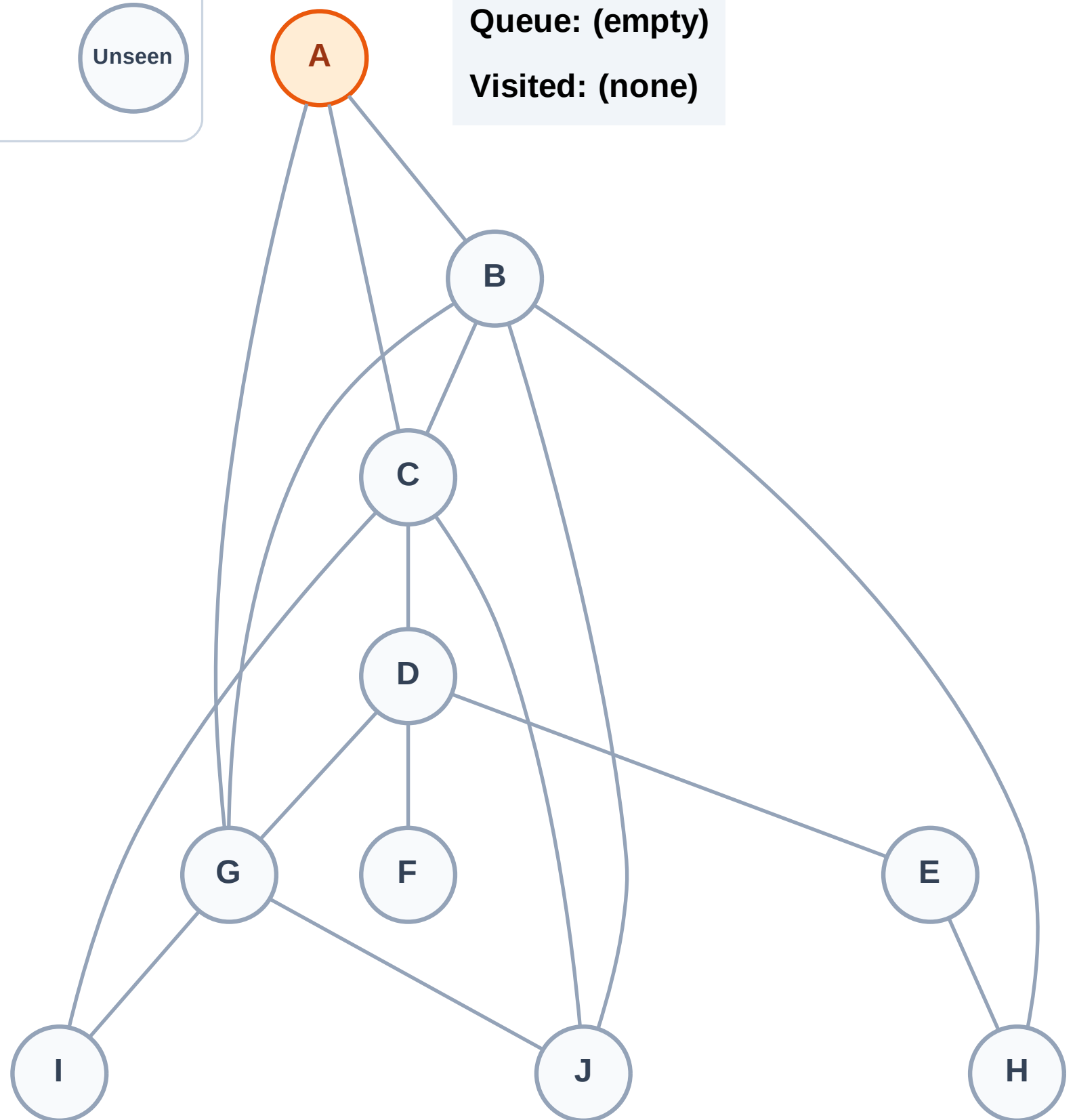
Step 2: Process node A

Legend



Queue: (empty)

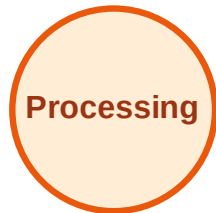
Visited: (none)



BFS Traversal

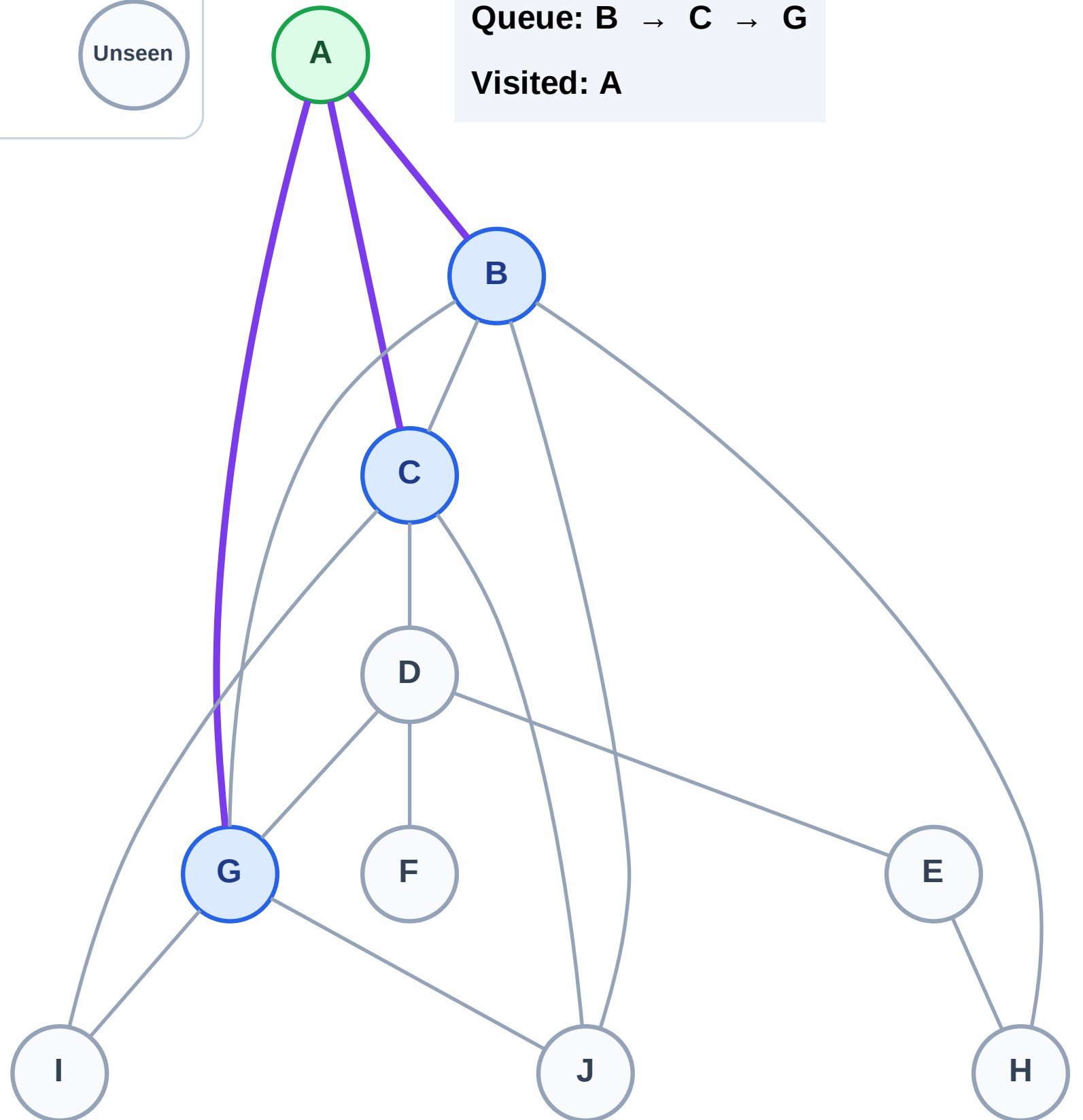
Step 3: Discover B, C, G from A

Legend



Queue: B → C → G

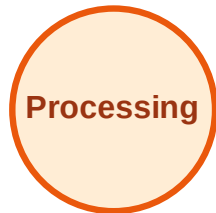
Visited: A



BFS Traversal

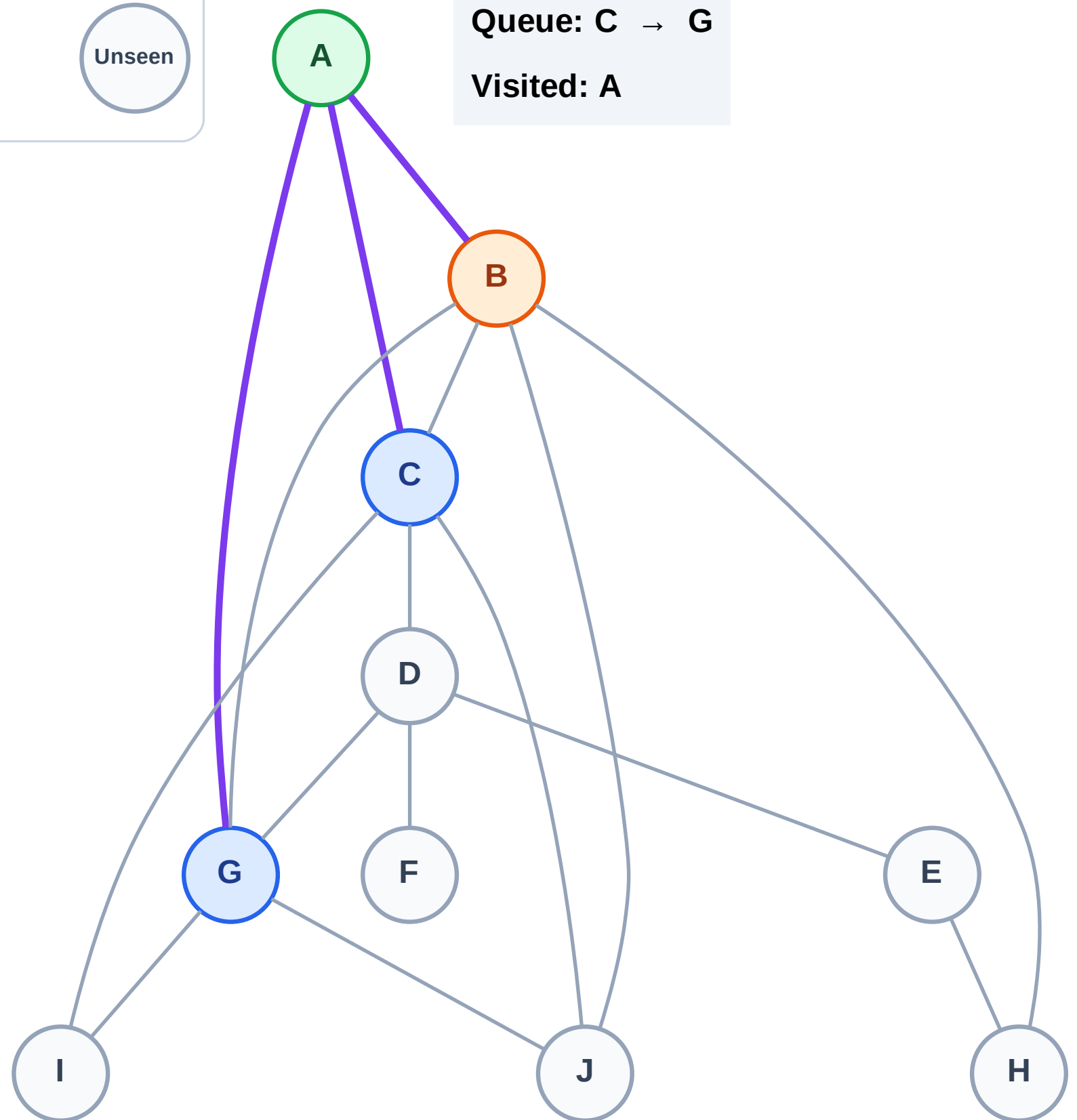
Step 4: Process node B

Legend



Queue: C → G

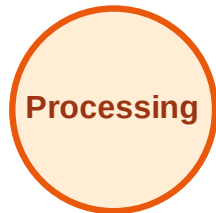
Visited: A



BFS Traversal

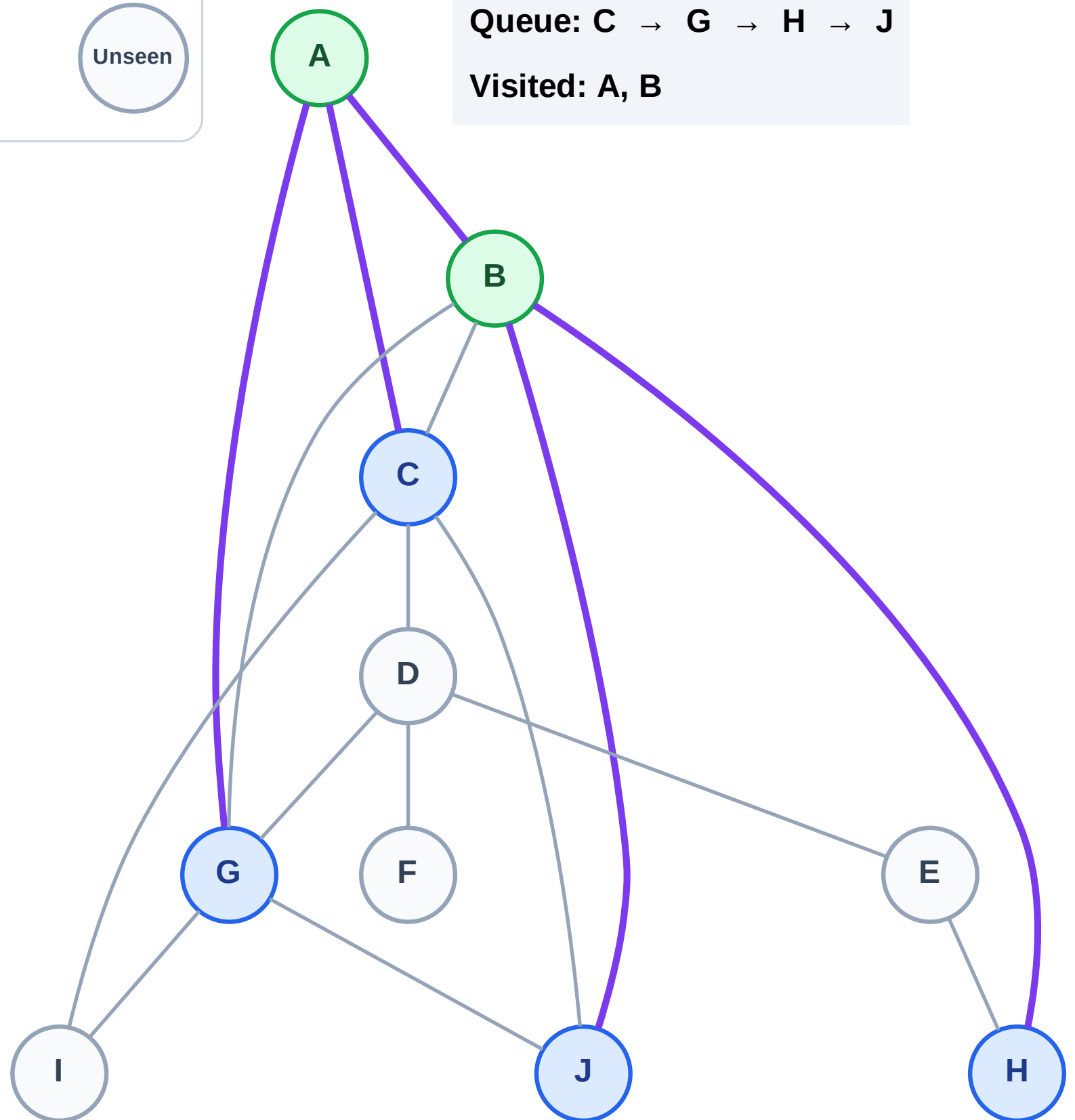
Step 5: Discover H, J from B

Legend



Queue: C → G → H → J

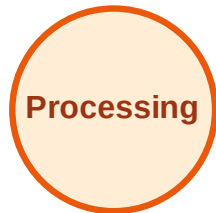
Visited: A, B



BFS Traversal

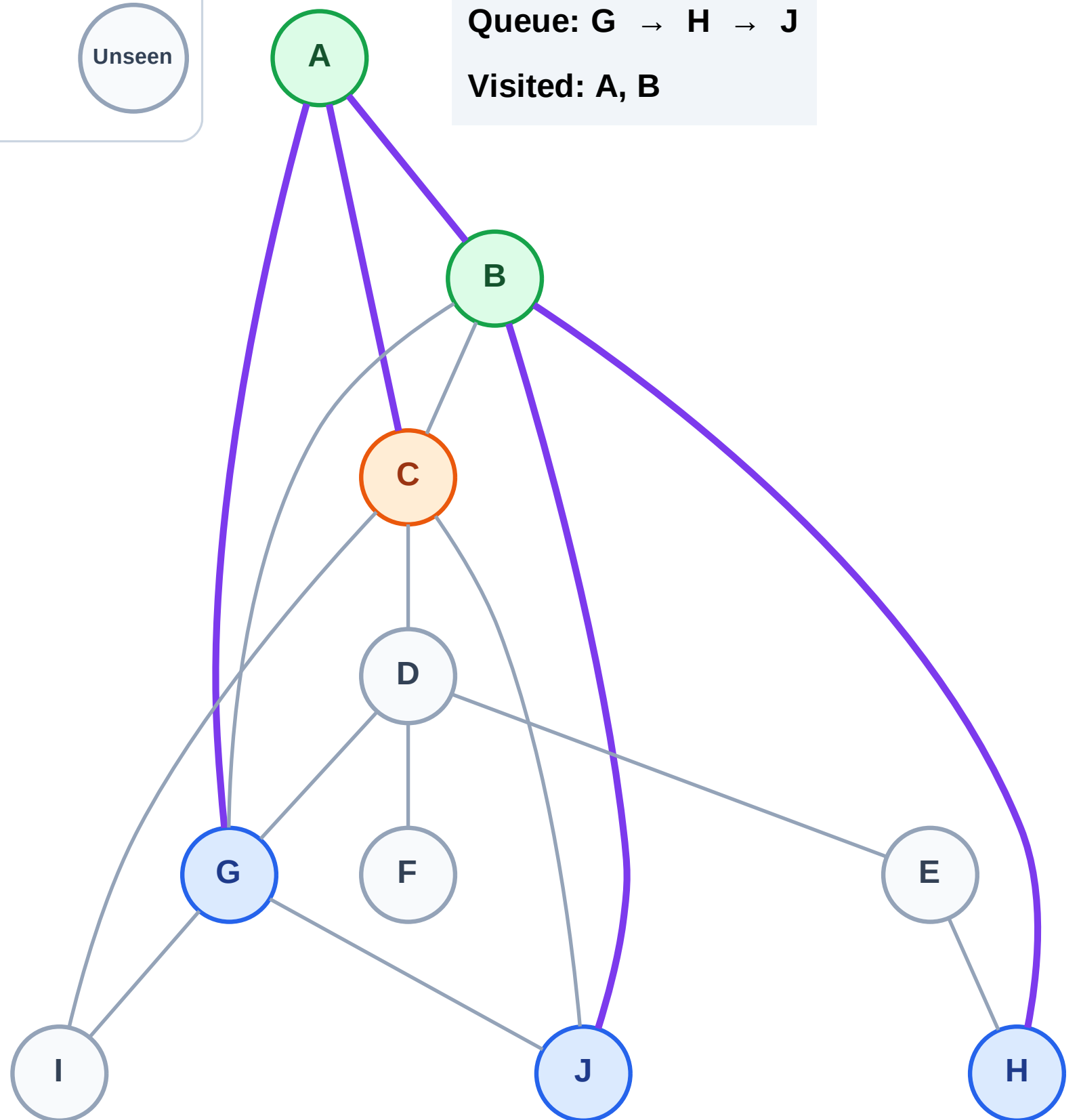
Step 6: Process node C

Legend



Queue: G → H → J

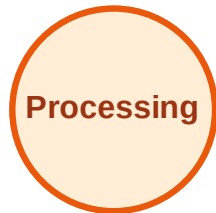
Visited: A, B



BFS Traversal

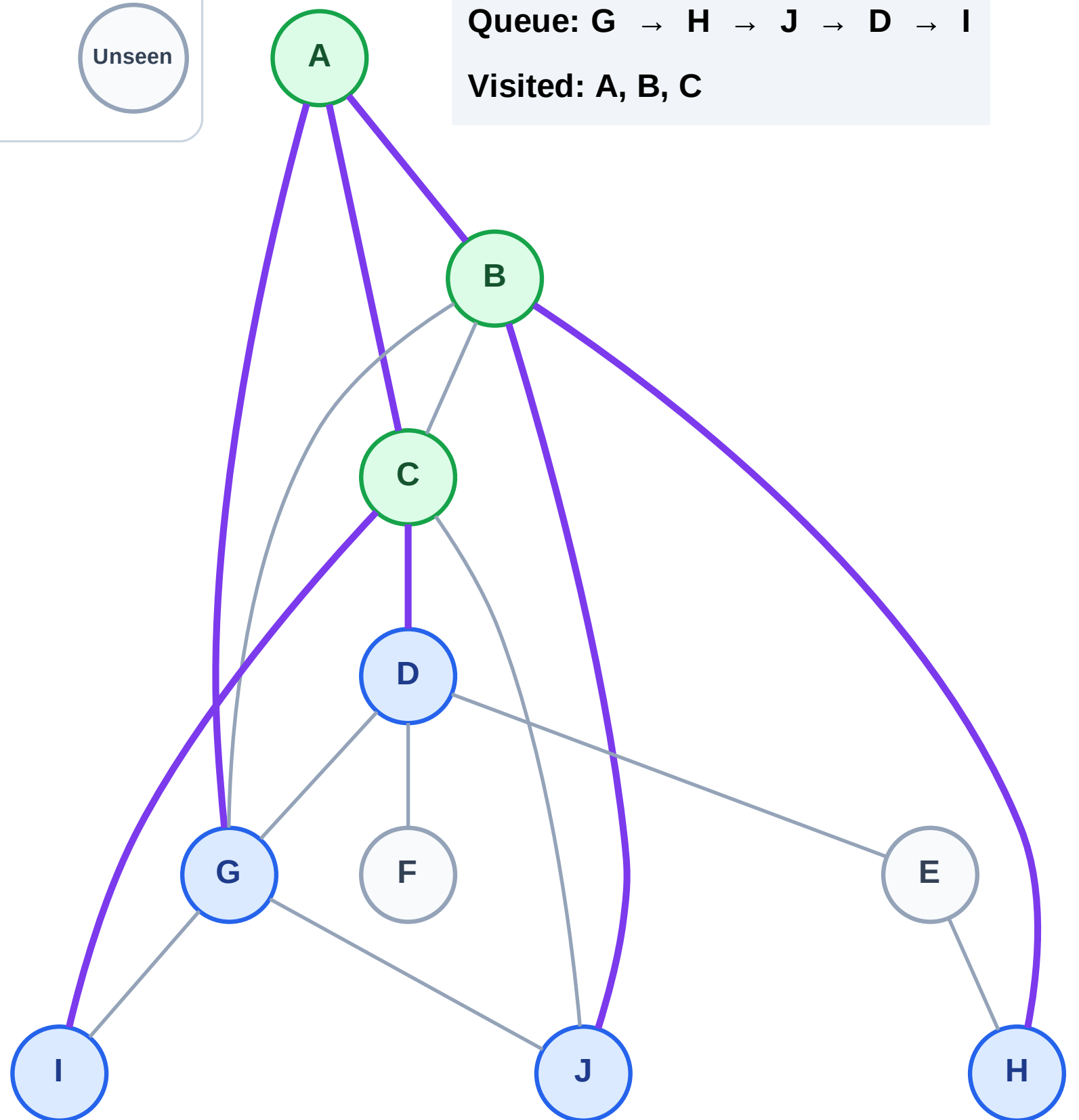
Step 7: Discover D, I from C

Legend



Queue: G → H → J → D → I

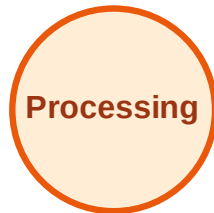
Visited: A, B, C



BFS Traversal

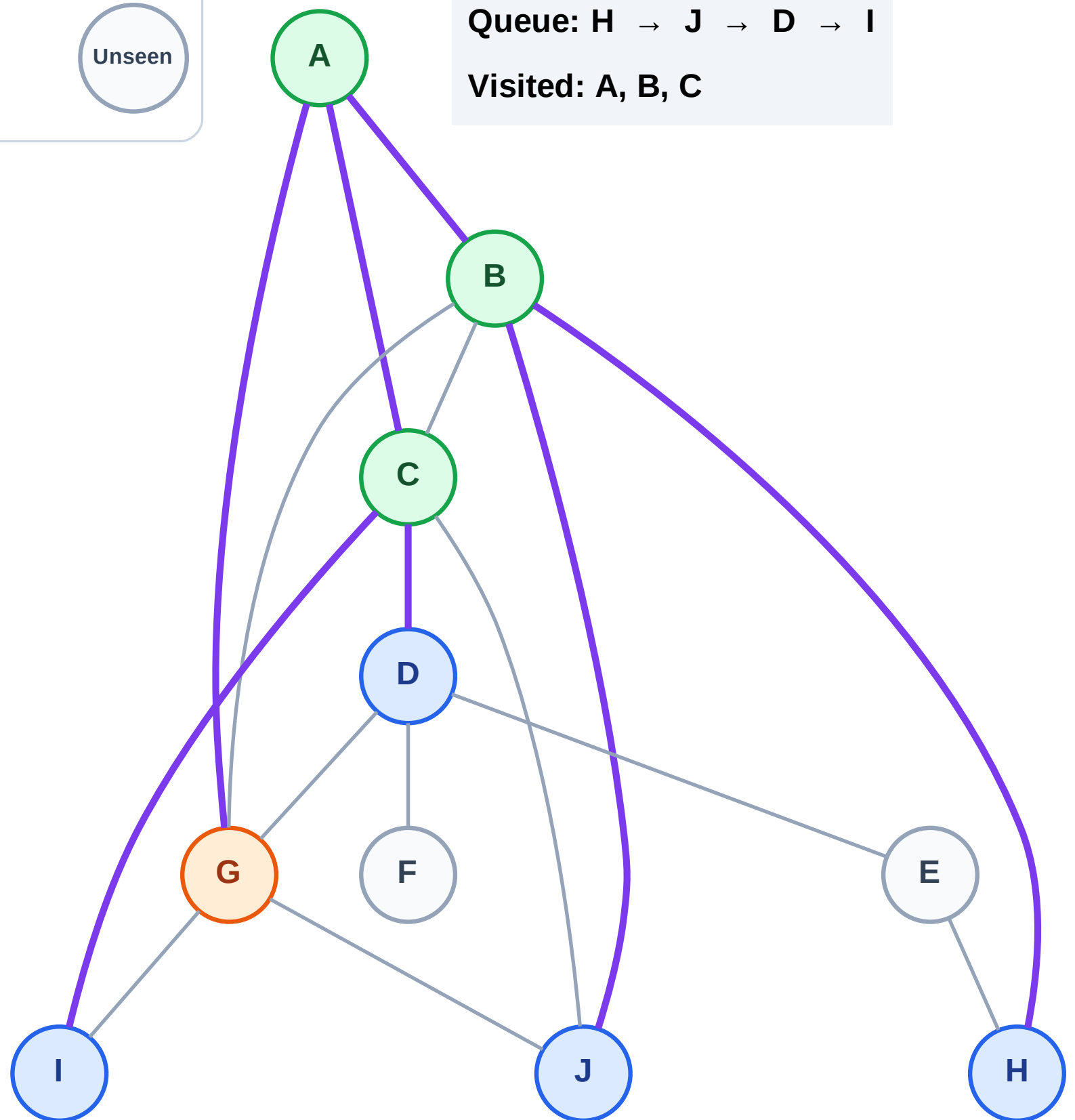
Step 8: Process node G

Legend



Queue: H → J → D → I

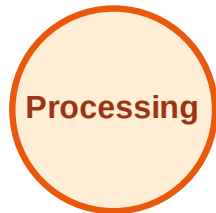
Visited: A, B, C



BFS Traversal

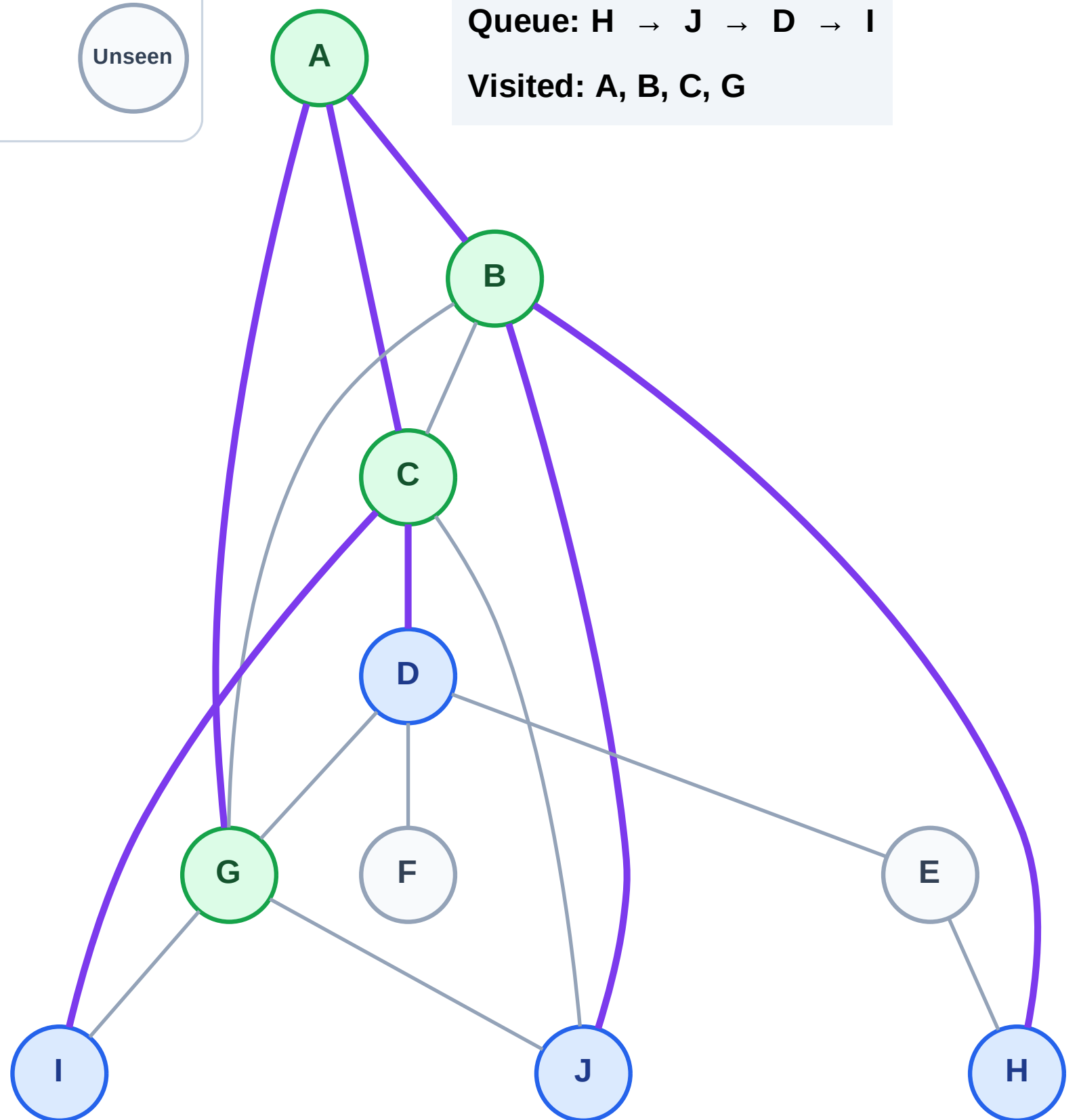
Step 9: No new neighbors from G

Legend



Queue: H → J → D → I

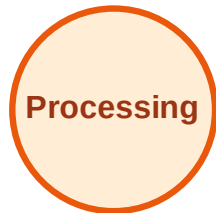
Visited: A, B, C, G



BFS Traversal

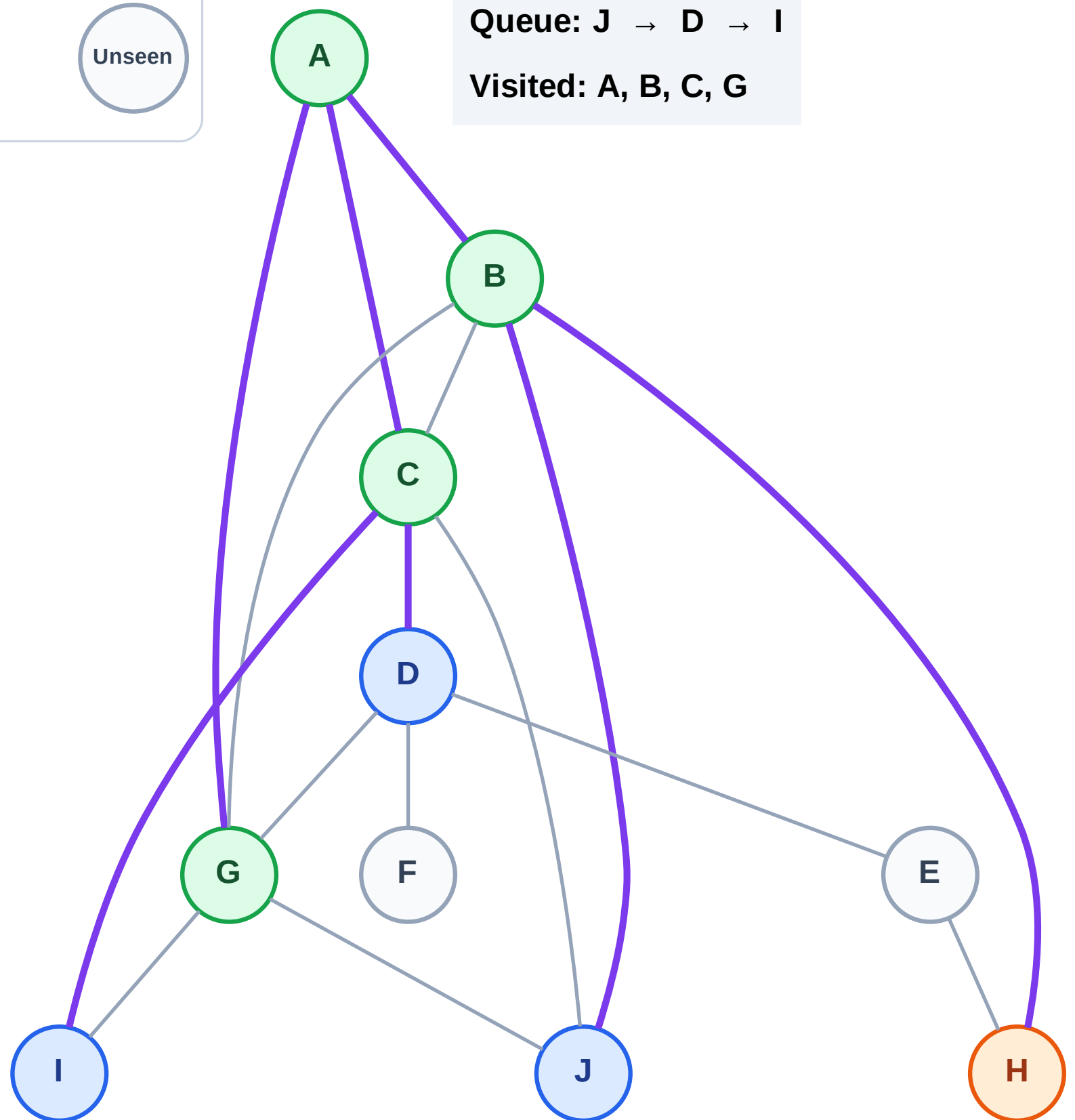
Step 10: Process node H

Legend



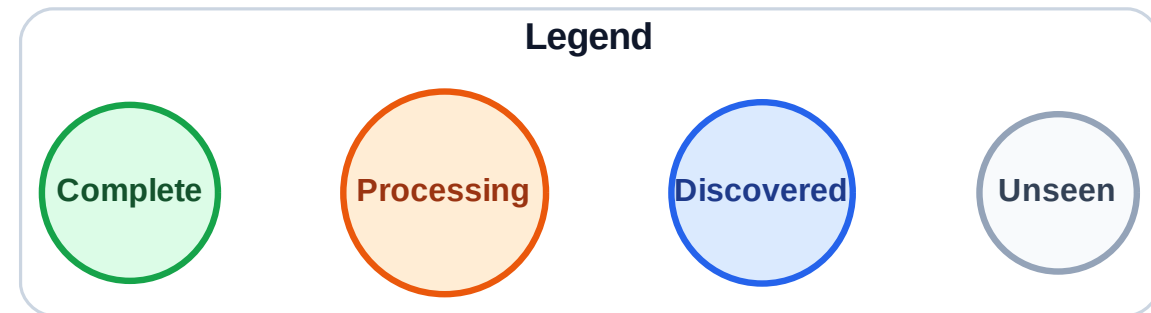
Queue: J → D → I

Visited: A, B, C, G

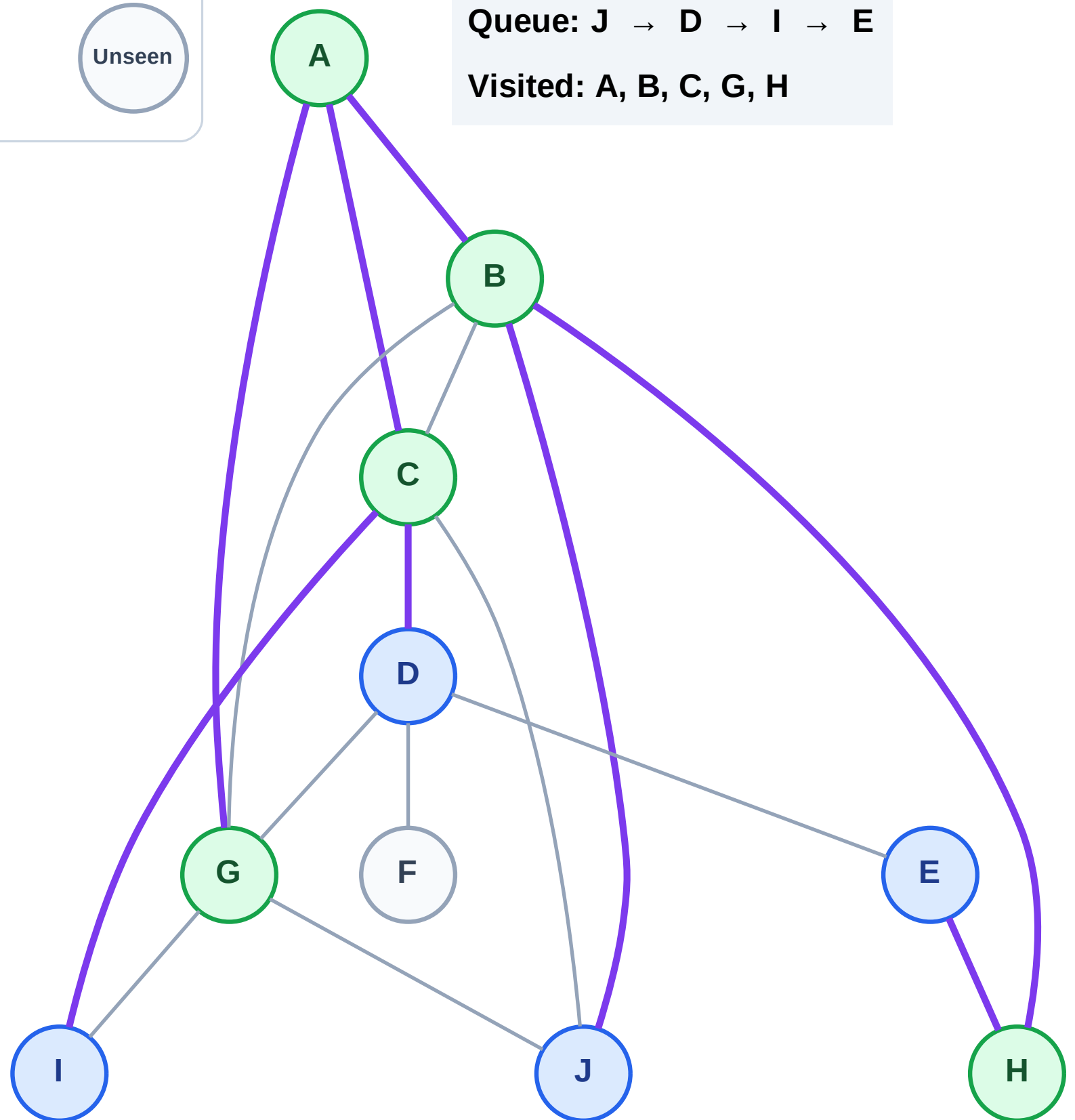


BFS Traversal

Step 11: Discover E from H



Queue: J → D → I → E
Visited: A, B, C, G, H



BFS Traversal

Step 12: Process node J

Legend

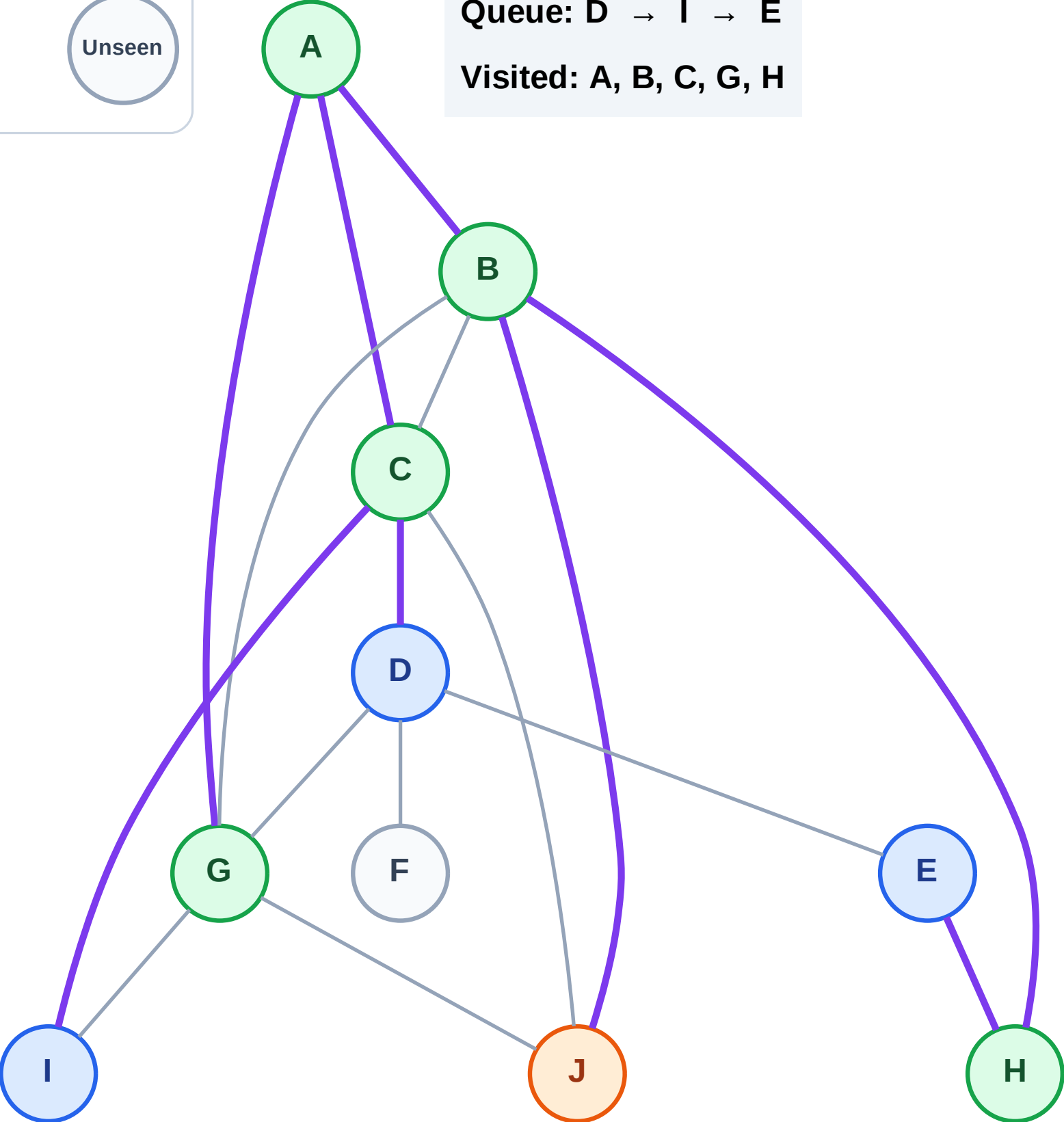
Complete

Processing

Discovered

Unseen

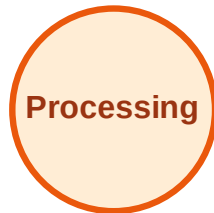
Queue: D → I → E
Visited: A, B, C, G, H



BFS Traversal

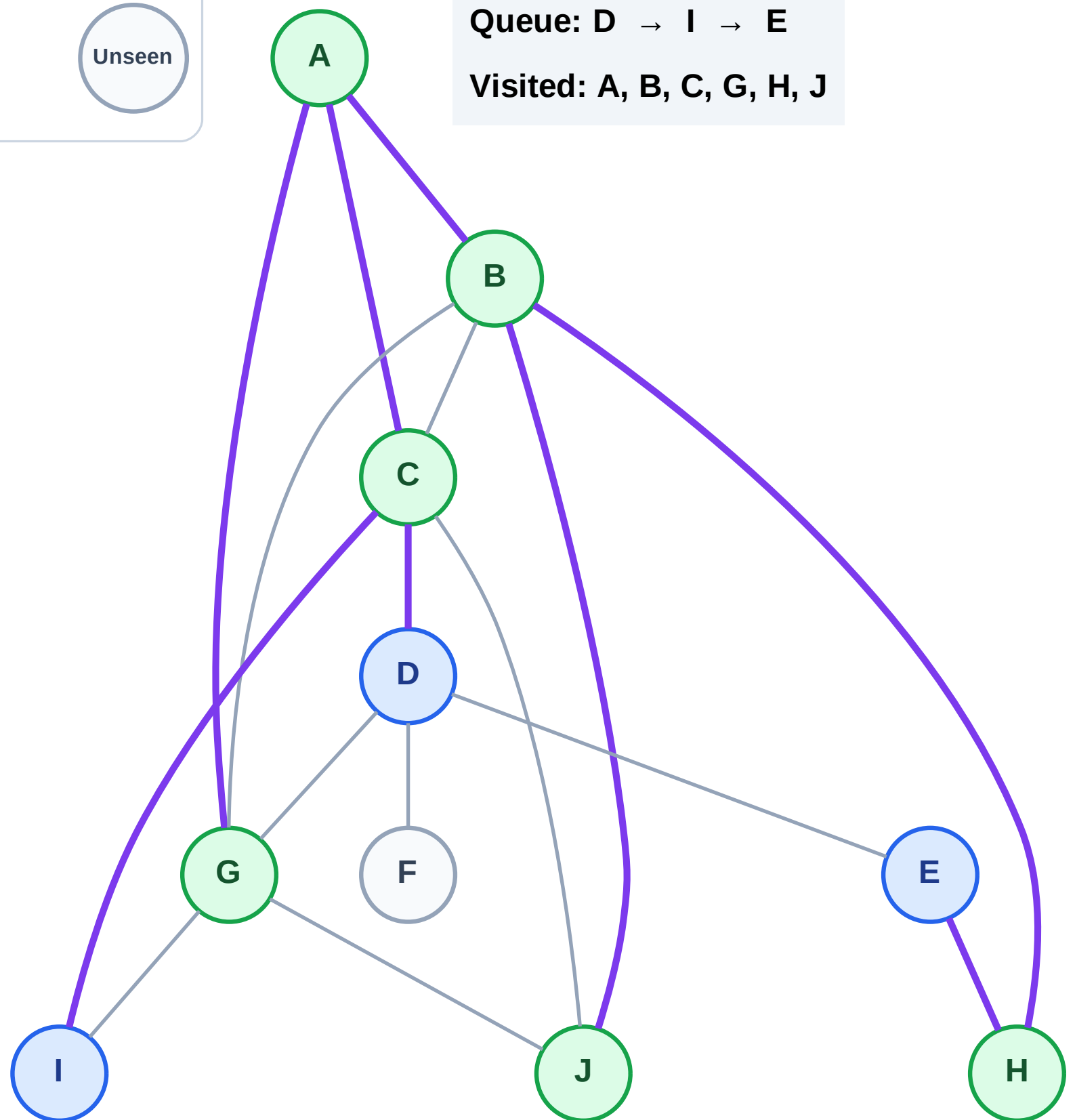
Step 13: No new neighbors from J

Legend



Queue: D → I → E

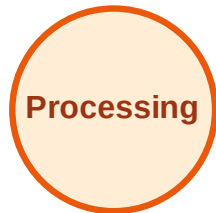
Visited: A, B, C, G, H, J



BFS Traversal

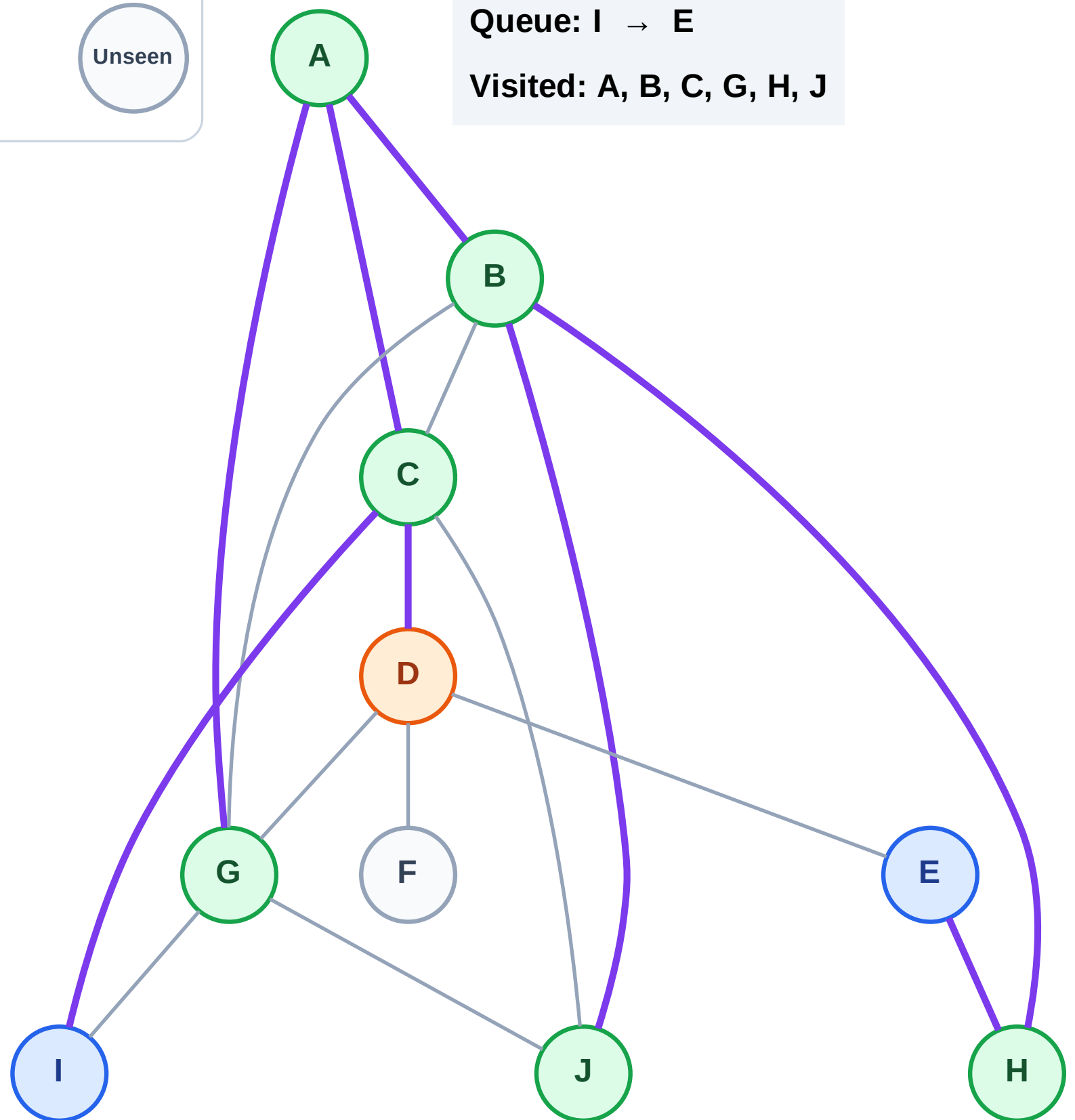
Step 14: Process node D

Legend



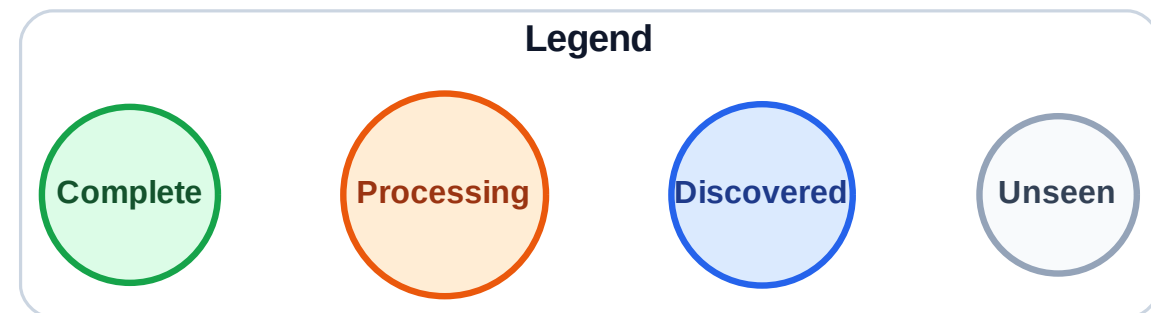
Queue: I → E

Visited: A, B, C, G, H, J

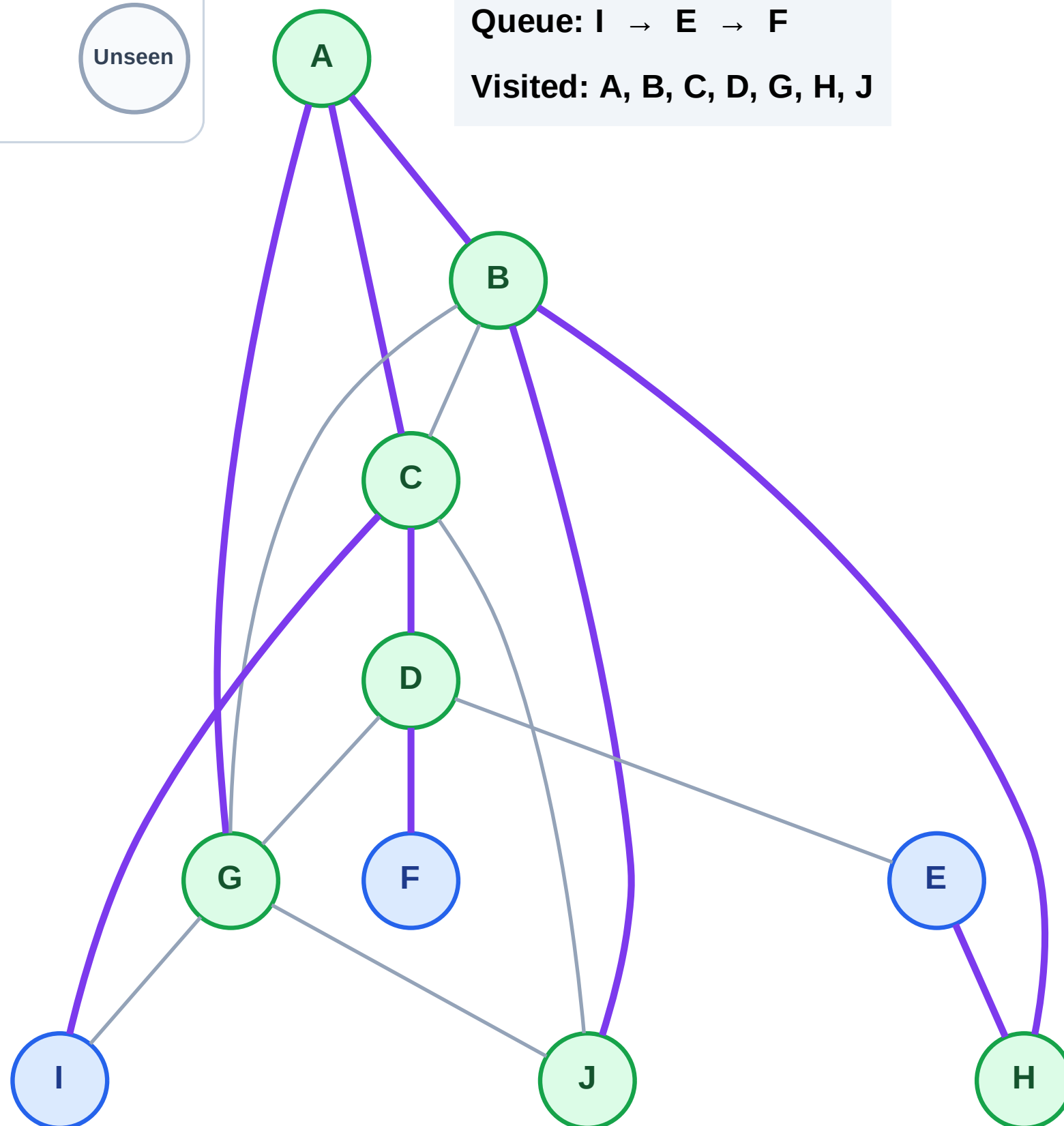


BFS Traversal

Step 15: Discover F from D



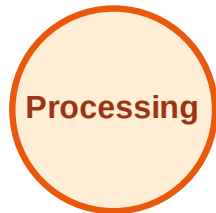
Queue: I → E → F
Visited: A, B, C, D, G, H, J



BFS Traversal

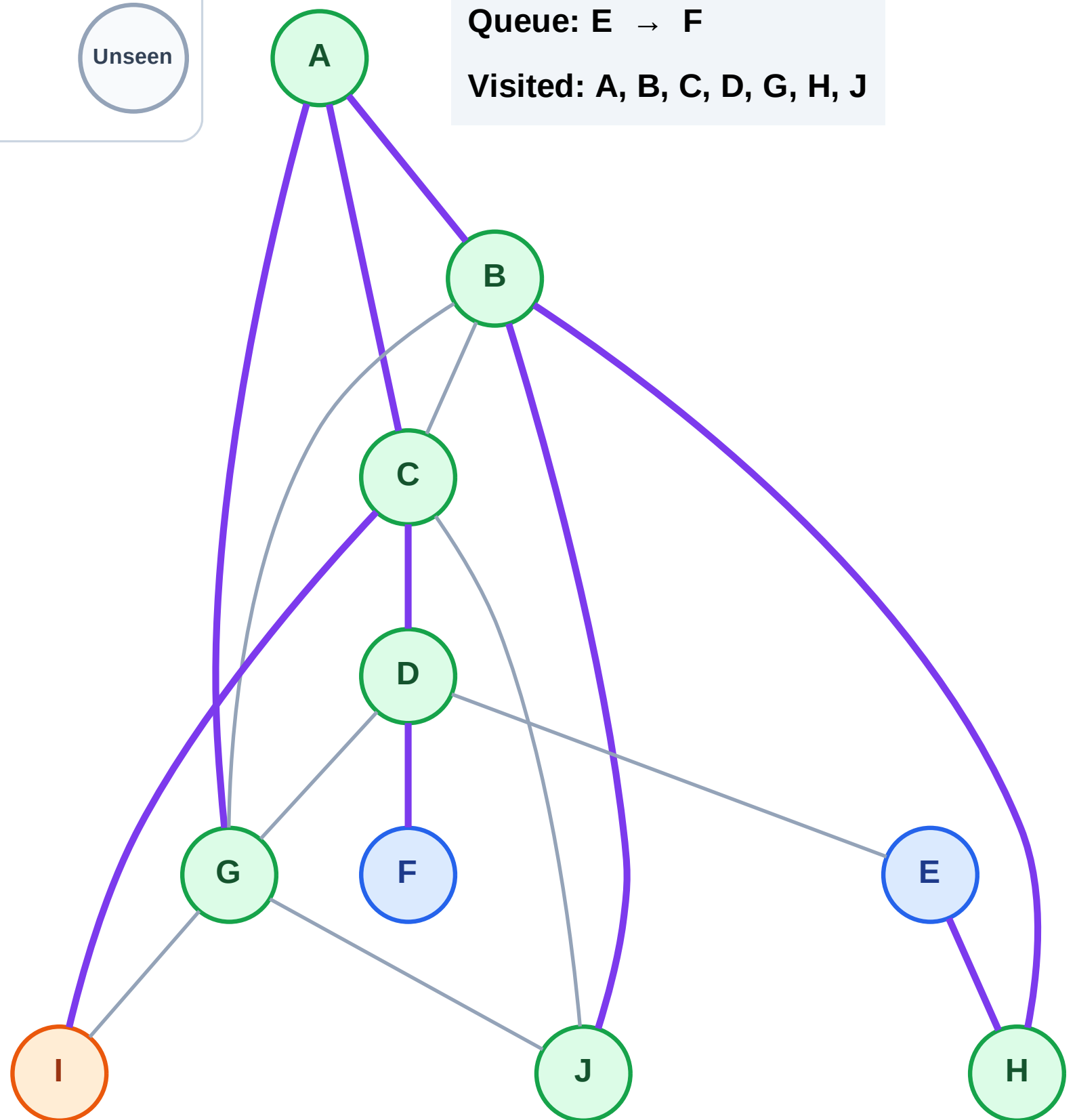
Step 16: Process node I

Legend



Queue: E → F

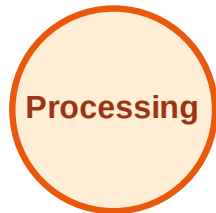
Visited: A, B, C, D, G, H, J



BFS Traversal

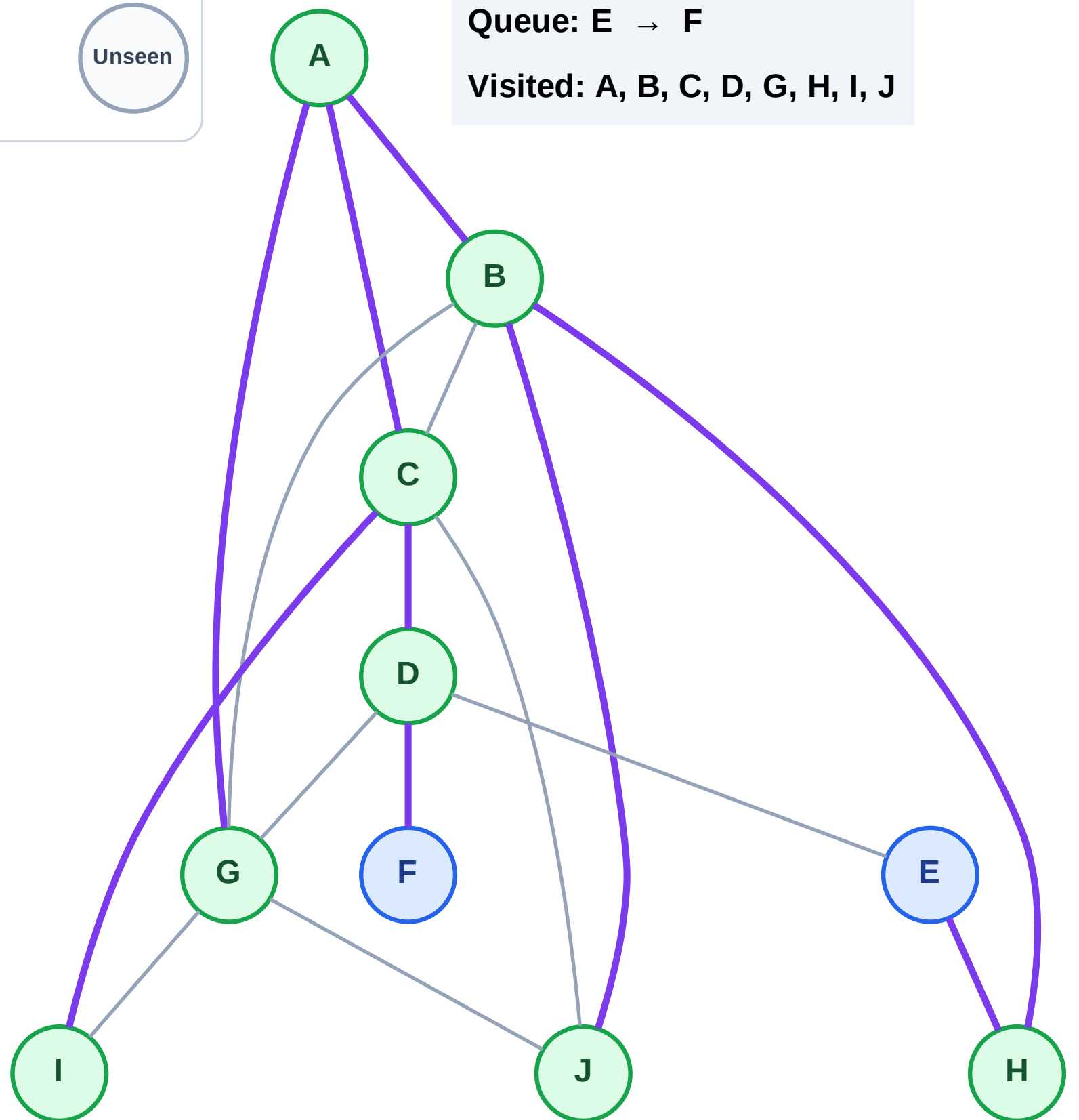
Step 17: No new neighbors from I

Legend



Queue: E → F

Visited: A, B, C, D, G, H, I, J



BFS Traversal

Step 18: Process node E

Legend

Complete

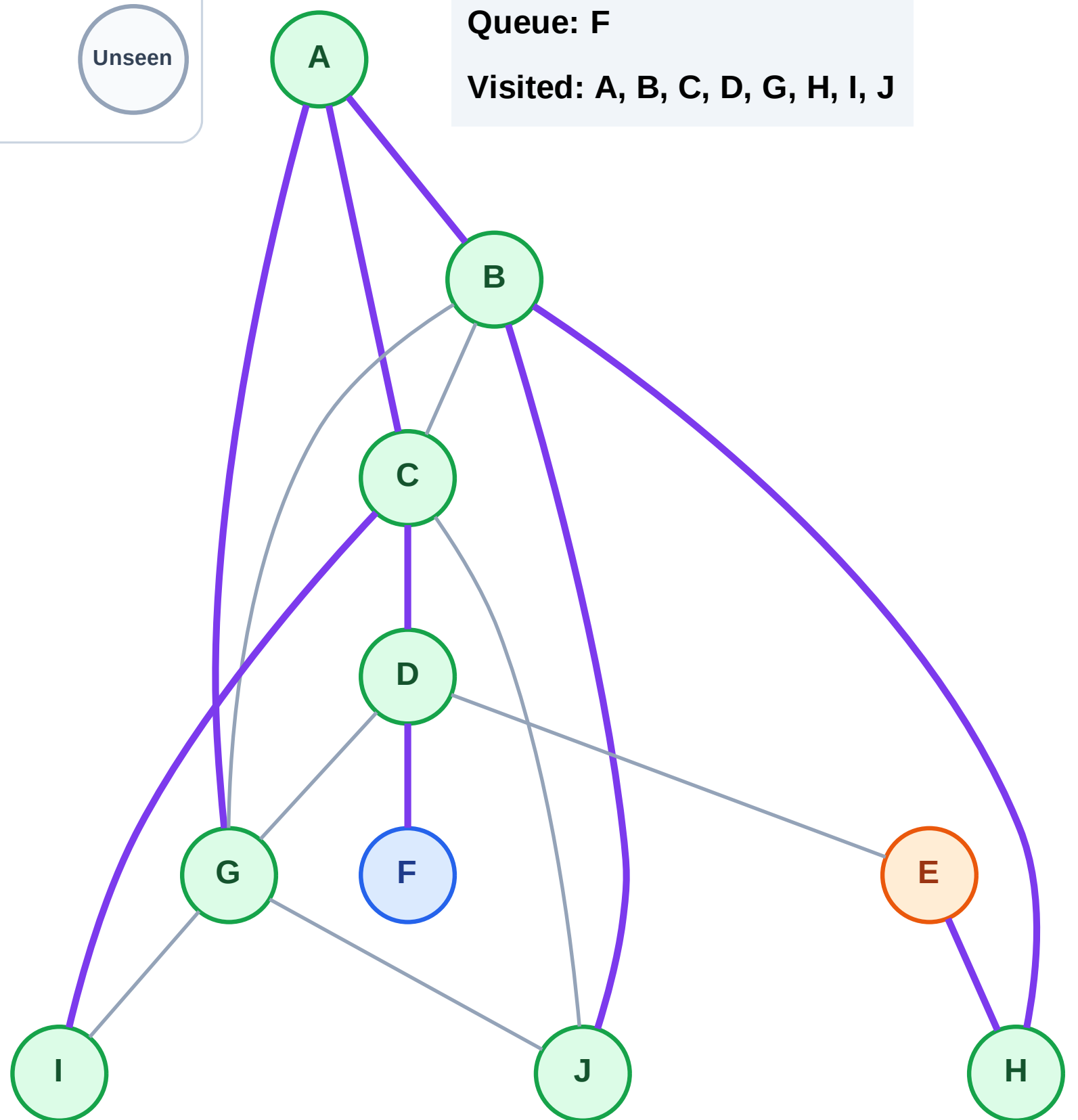
Processing

Discovered

Unseen

Queue: F

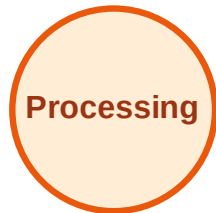
Visited: A, B, C, D, G, H, I, J



BFS Traversal

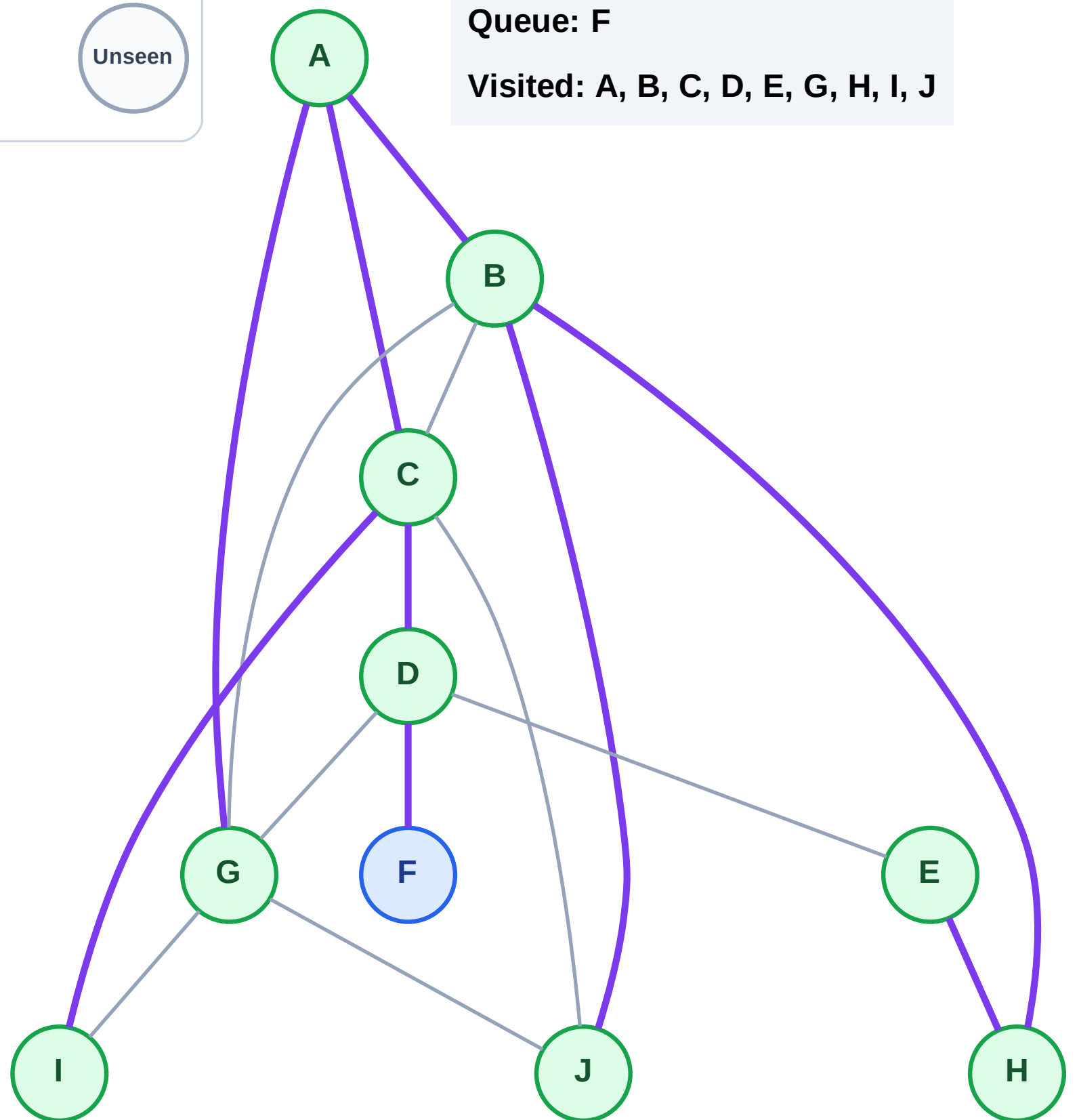
Step 19: No new neighbors from E

Legend



Queue: F

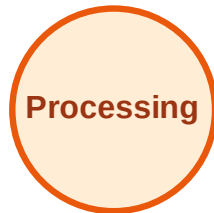
Visited: A, B, C, D, E, G, H, I, J



BFS Traversal

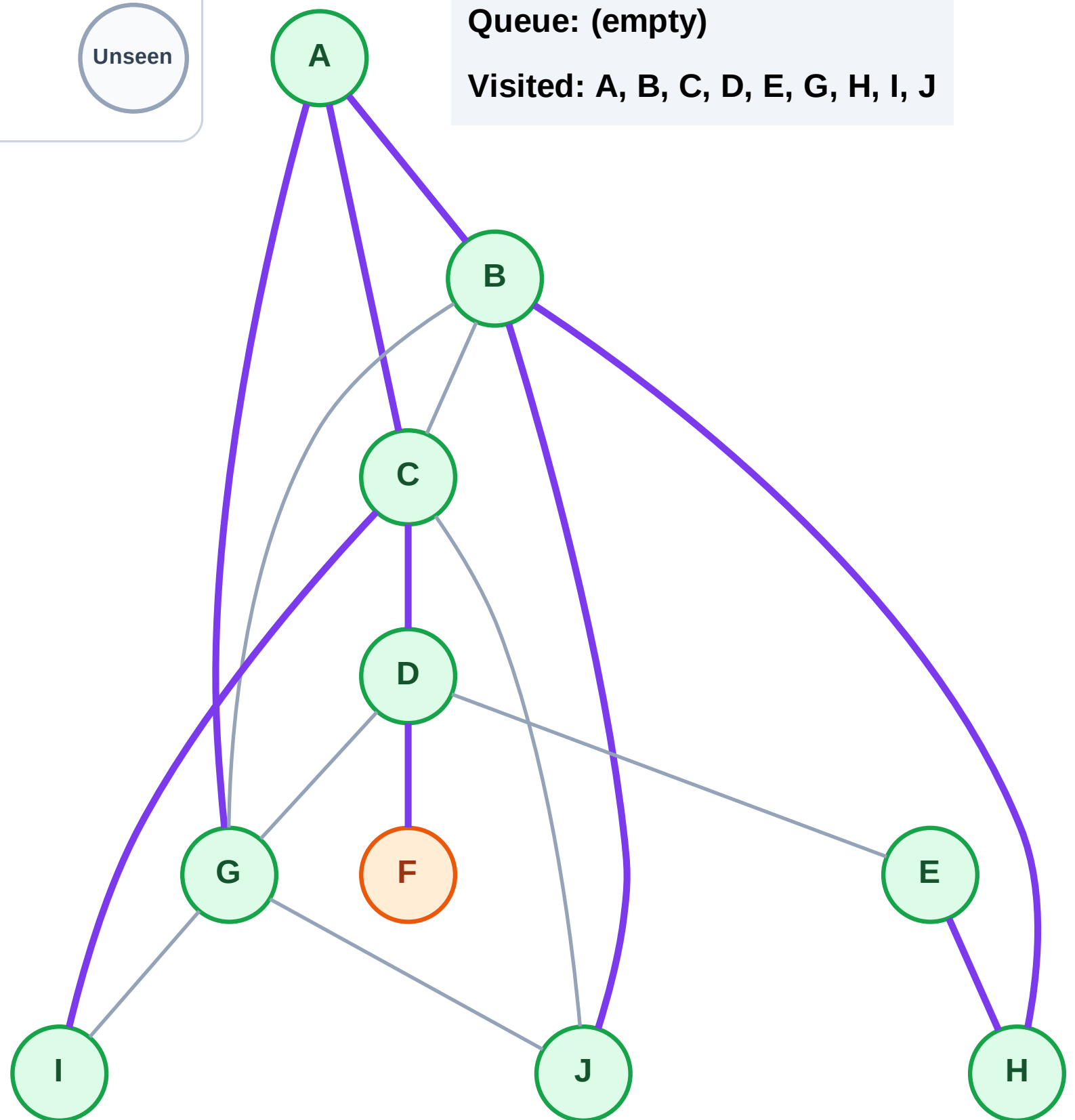
Step 20: Process node F

Legend



Queue: (empty)

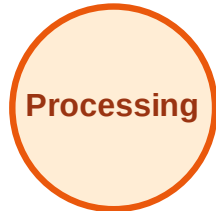
Visited: A, B, C, D, E, G, H, I, J



BFS Traversal

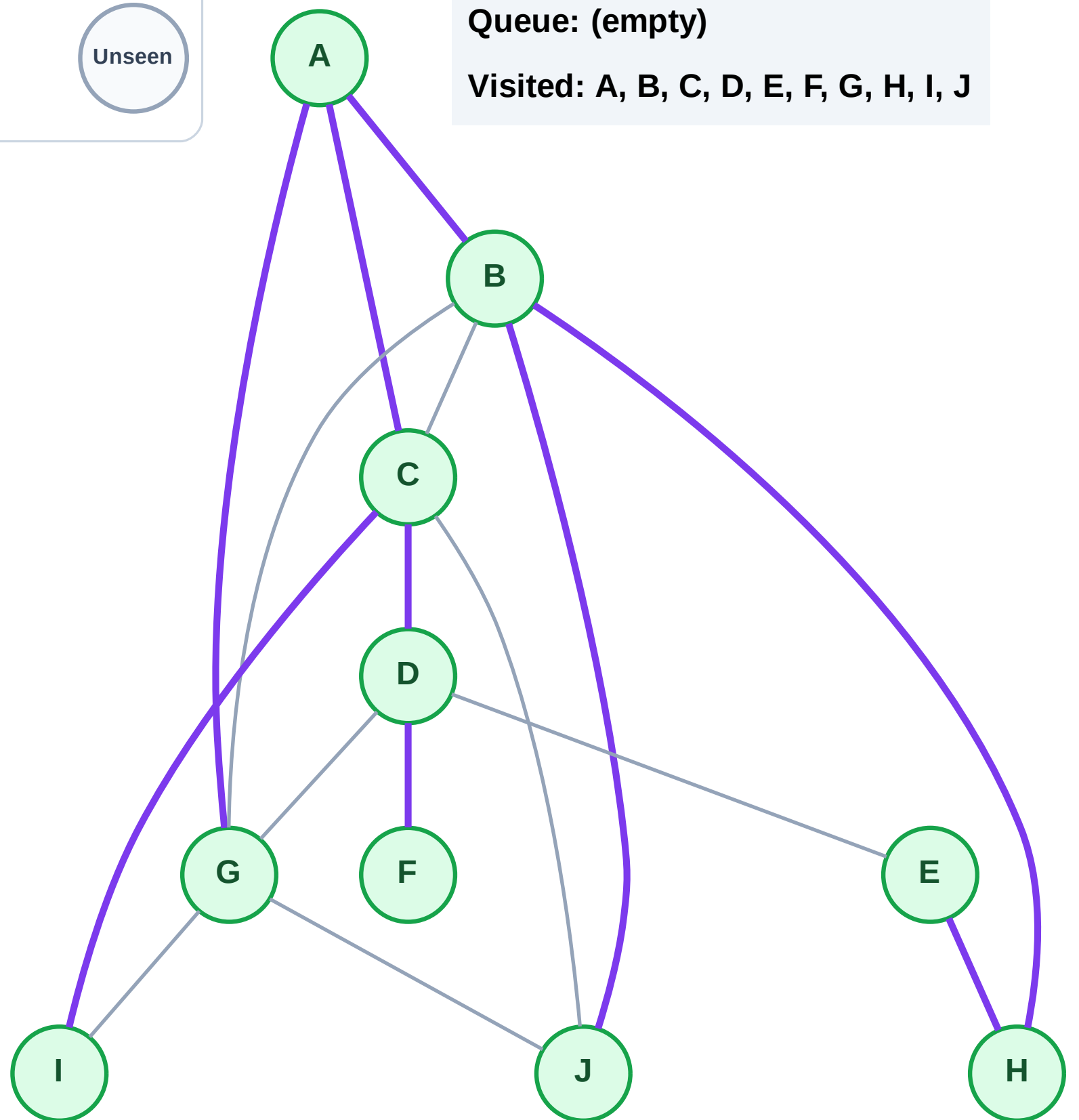
Step 21: No new neighbors from F

Legend



Queue: (empty)

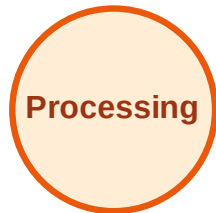
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

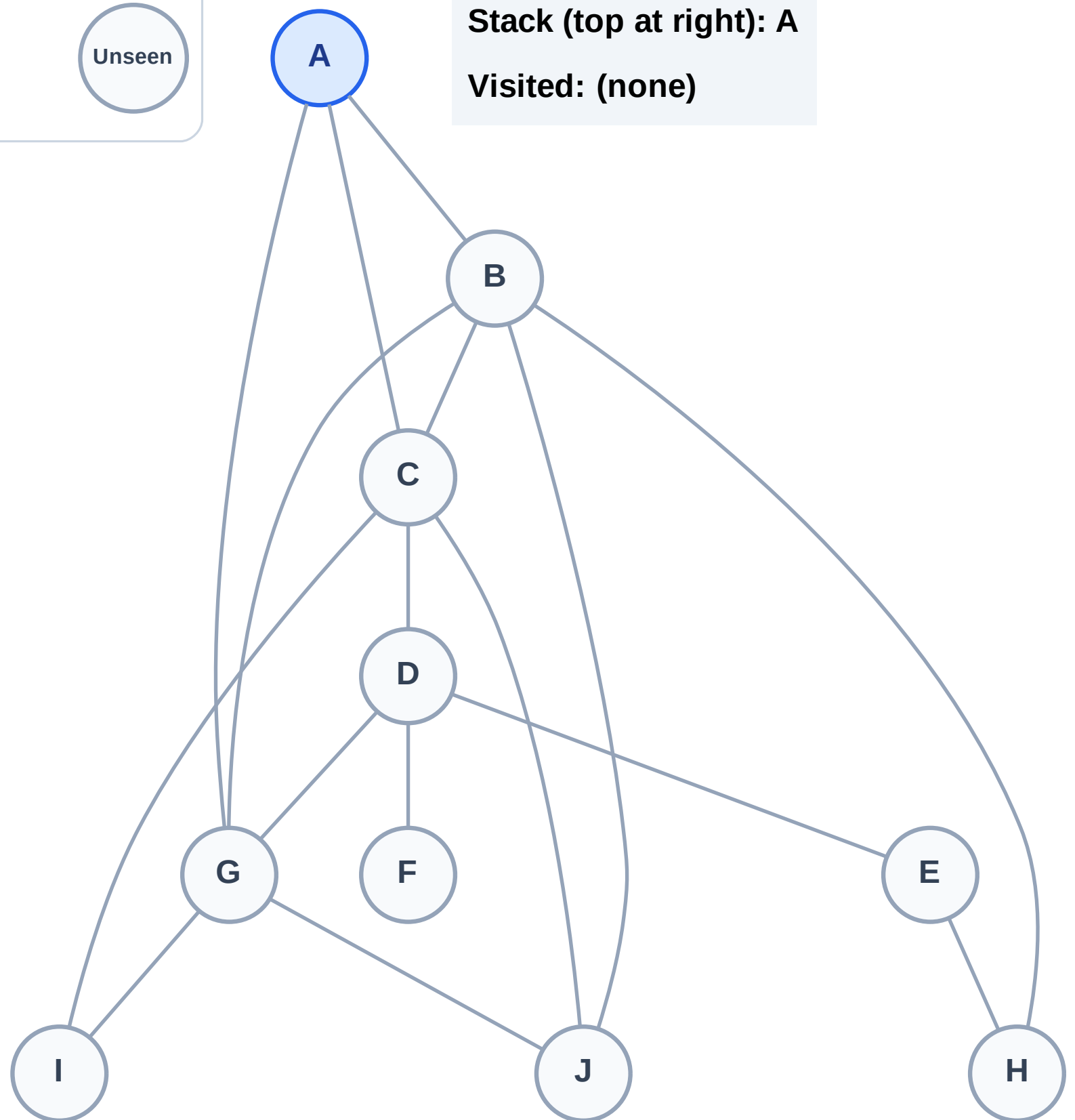
Step 1: Start at A

Legend



Stack (top at right): A

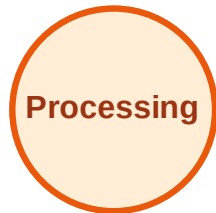
Visited: (none)



DFS Traversal

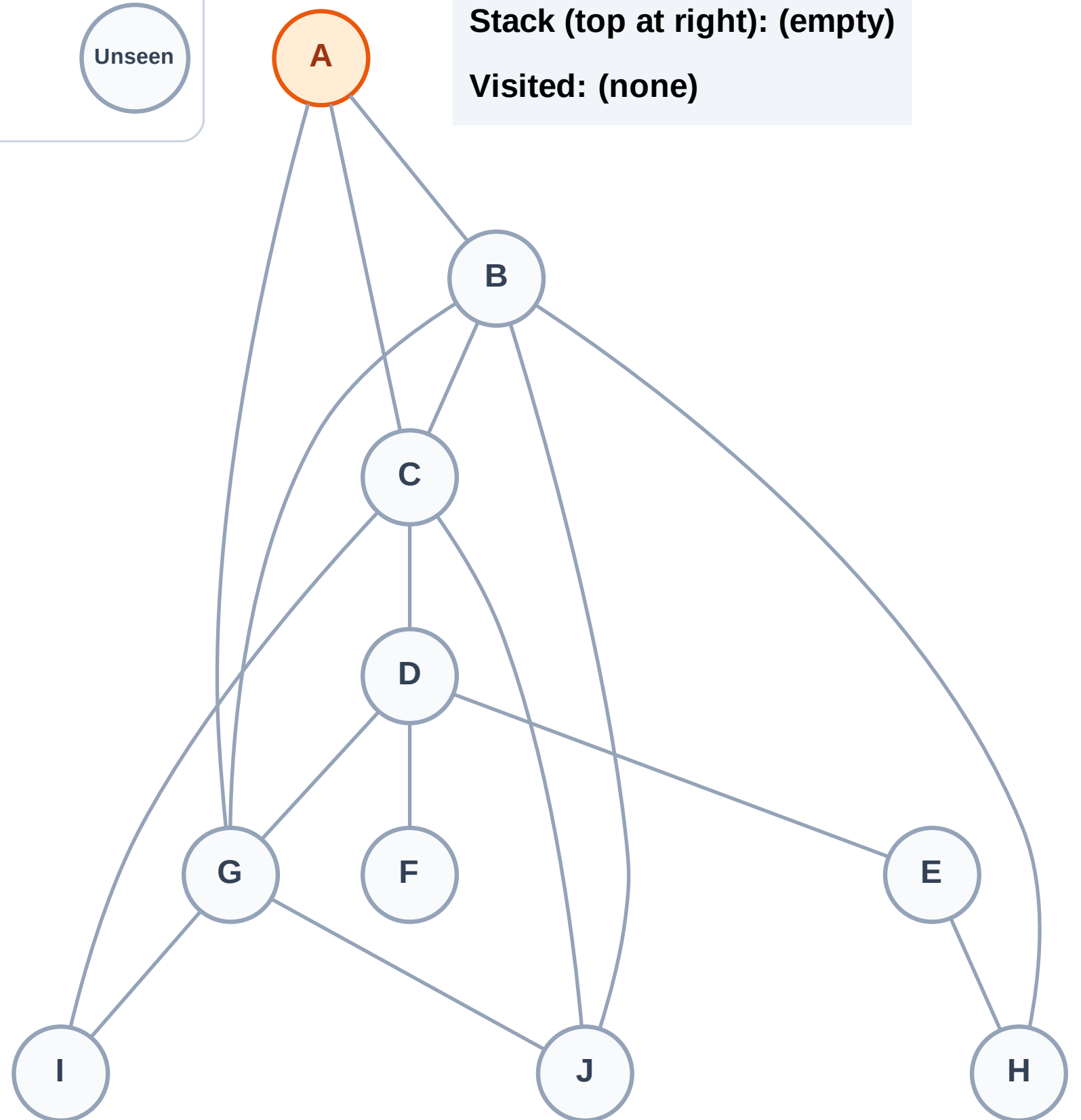
Step 2: Process node A

Legend



Stack (top at right): (empty)

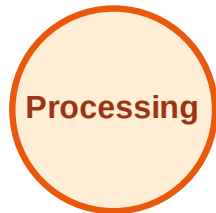
Visited: (none)



DFS Traversal

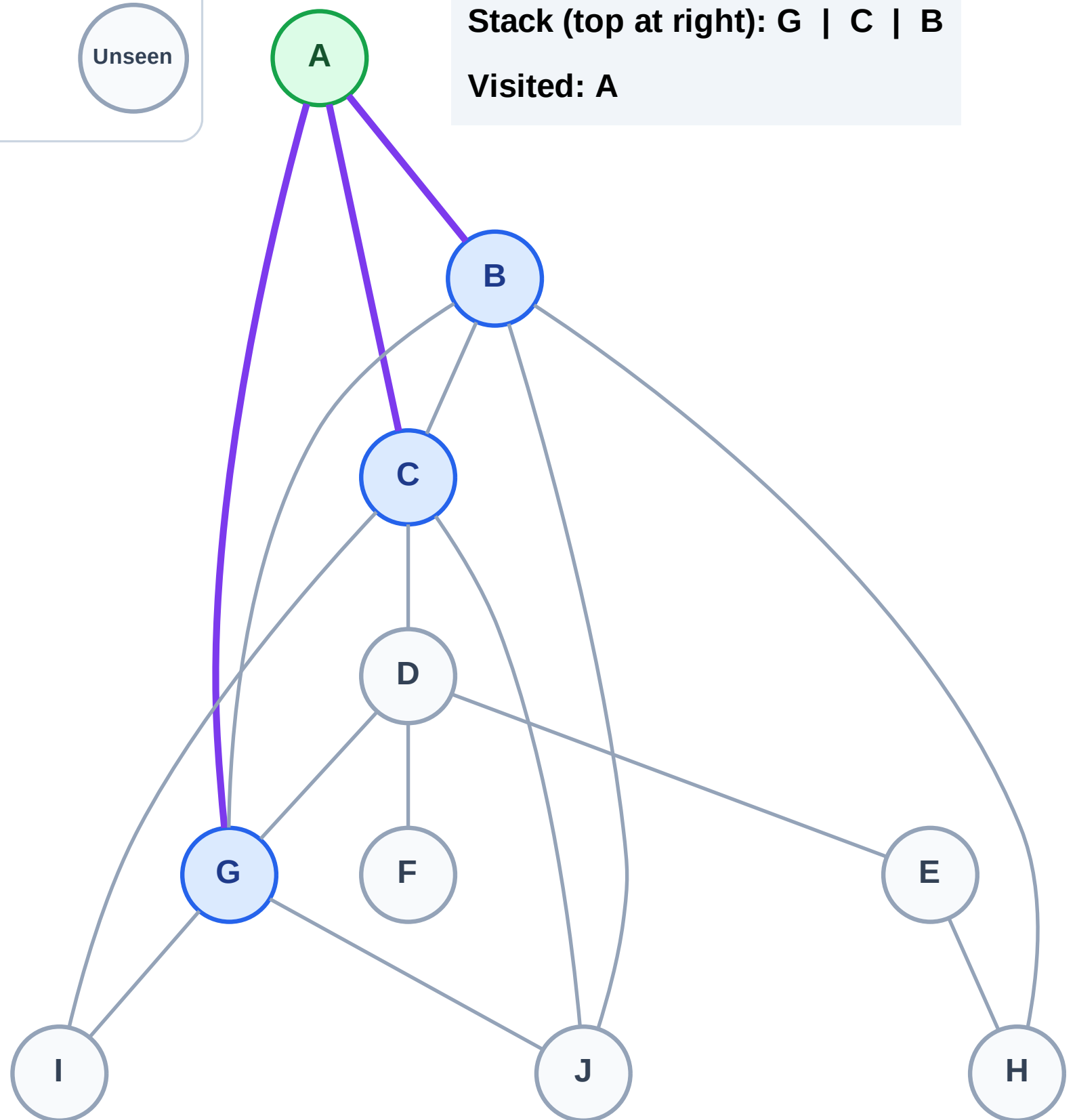
Step 3: Discover G, C, B from A

Legend



Stack (top at right): G | C | B

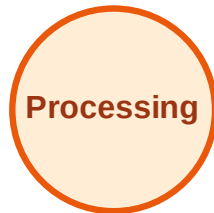
Visited: A



DFS Traversal

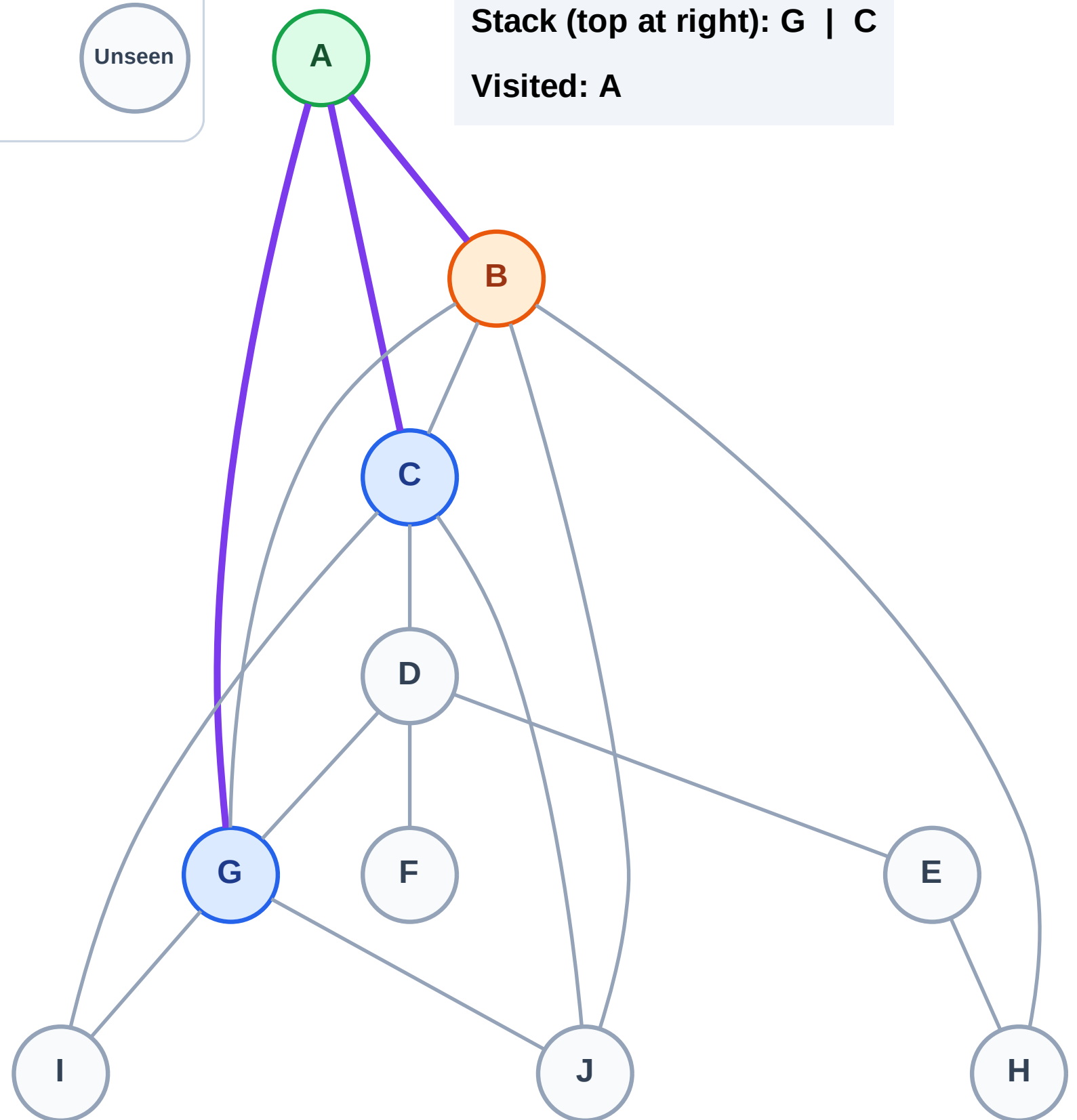
Step 4: Process node B

Legend



Stack (top at right): G | C

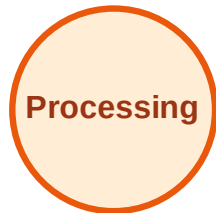
Visited: A



DFS Traversal

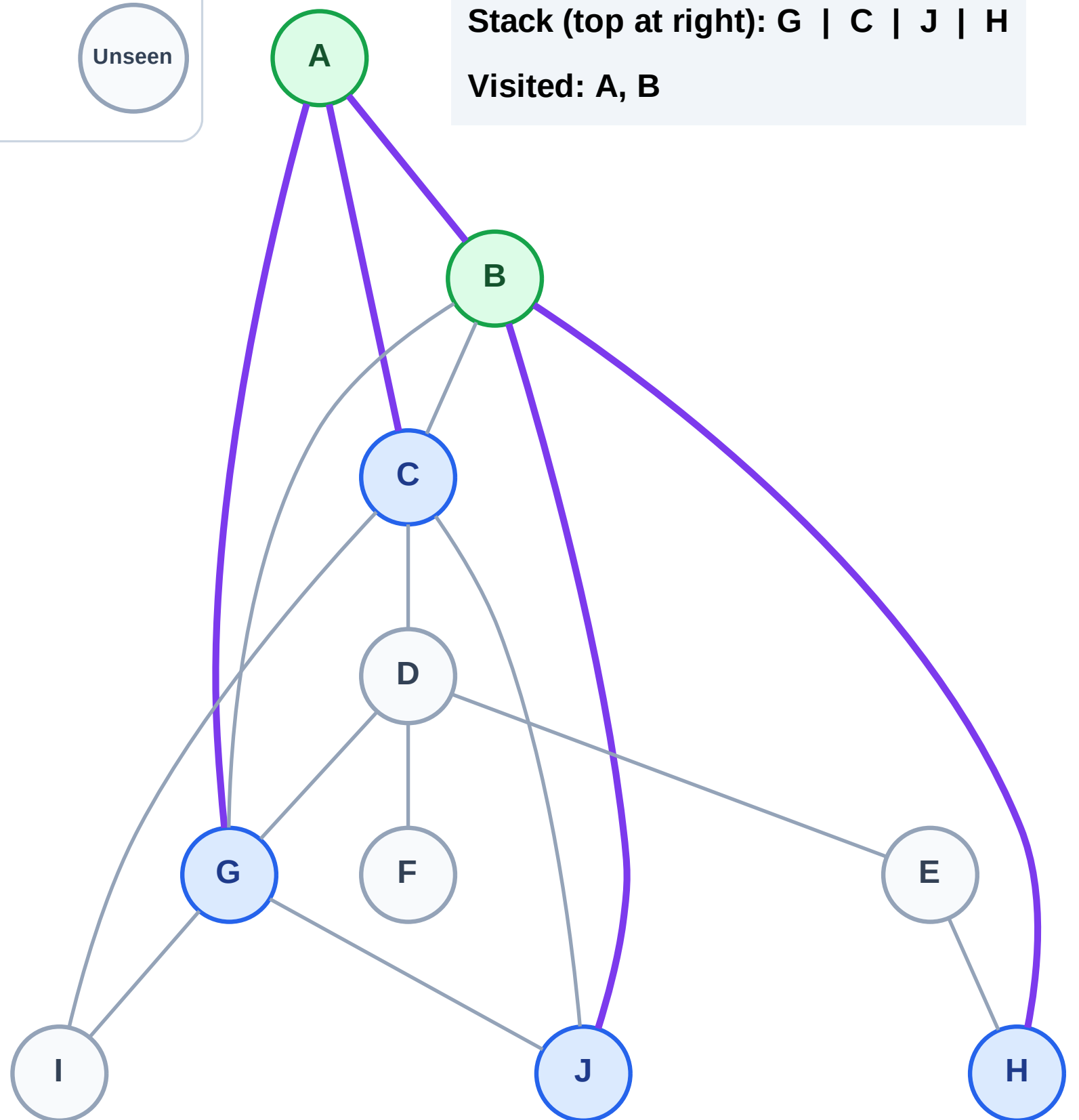
Step 5: Discover J, H from B

Legend



Stack (top at right): G | C | J | H

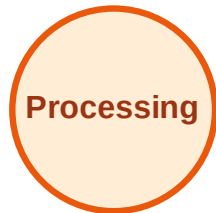
Visited: A, B



DFS Traversal

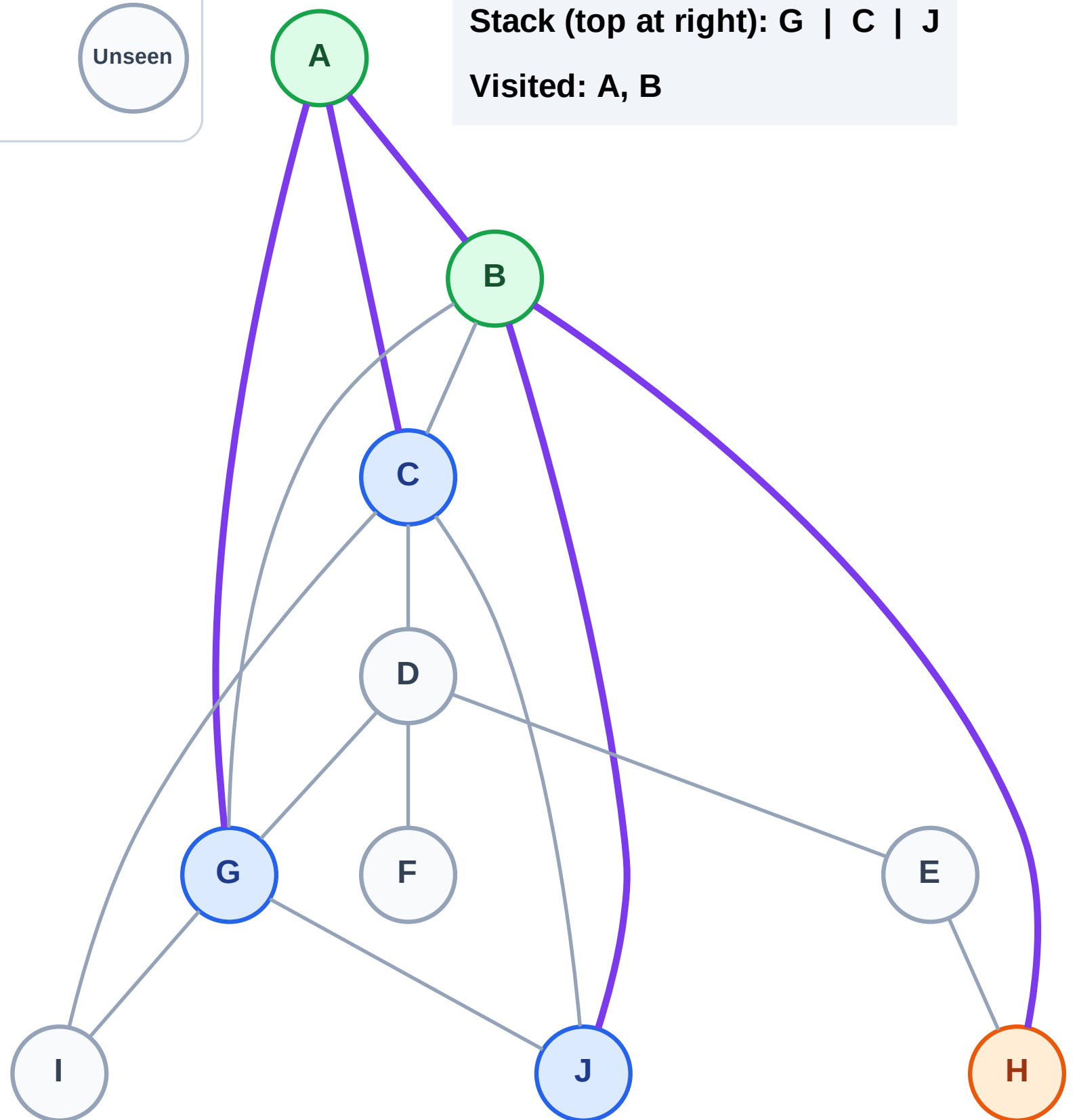
Step 6: Process node H

Legend



Stack (top at right): G | C | J

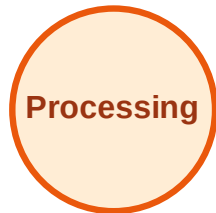
Visited: A, B



DFS Traversal

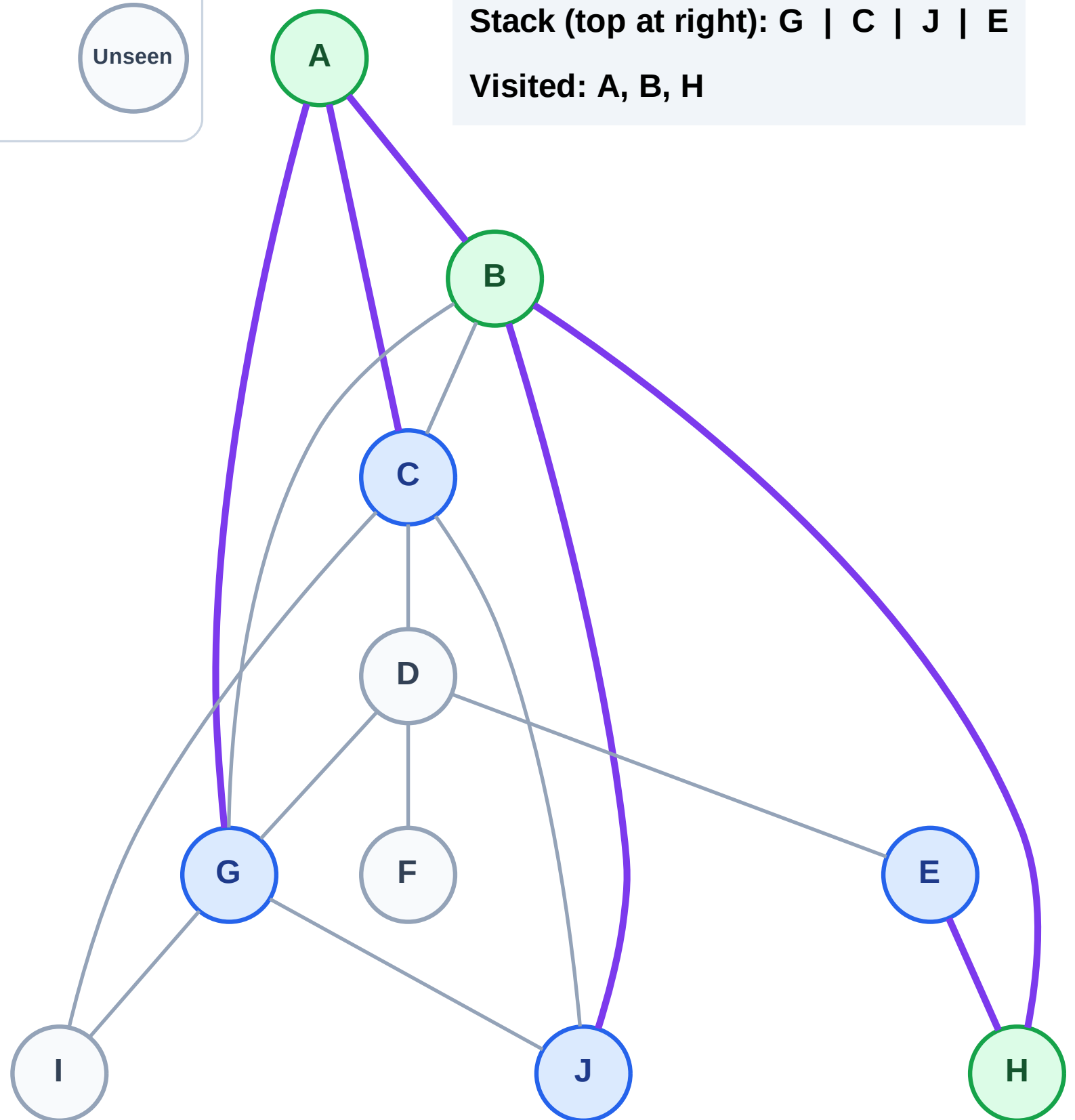
Step 7: Discover E from H

Legend



Stack (top at right): G | C | J | E

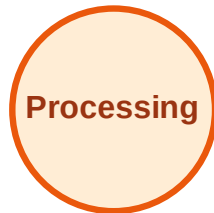
Visited: A, B, H



DFS Traversal

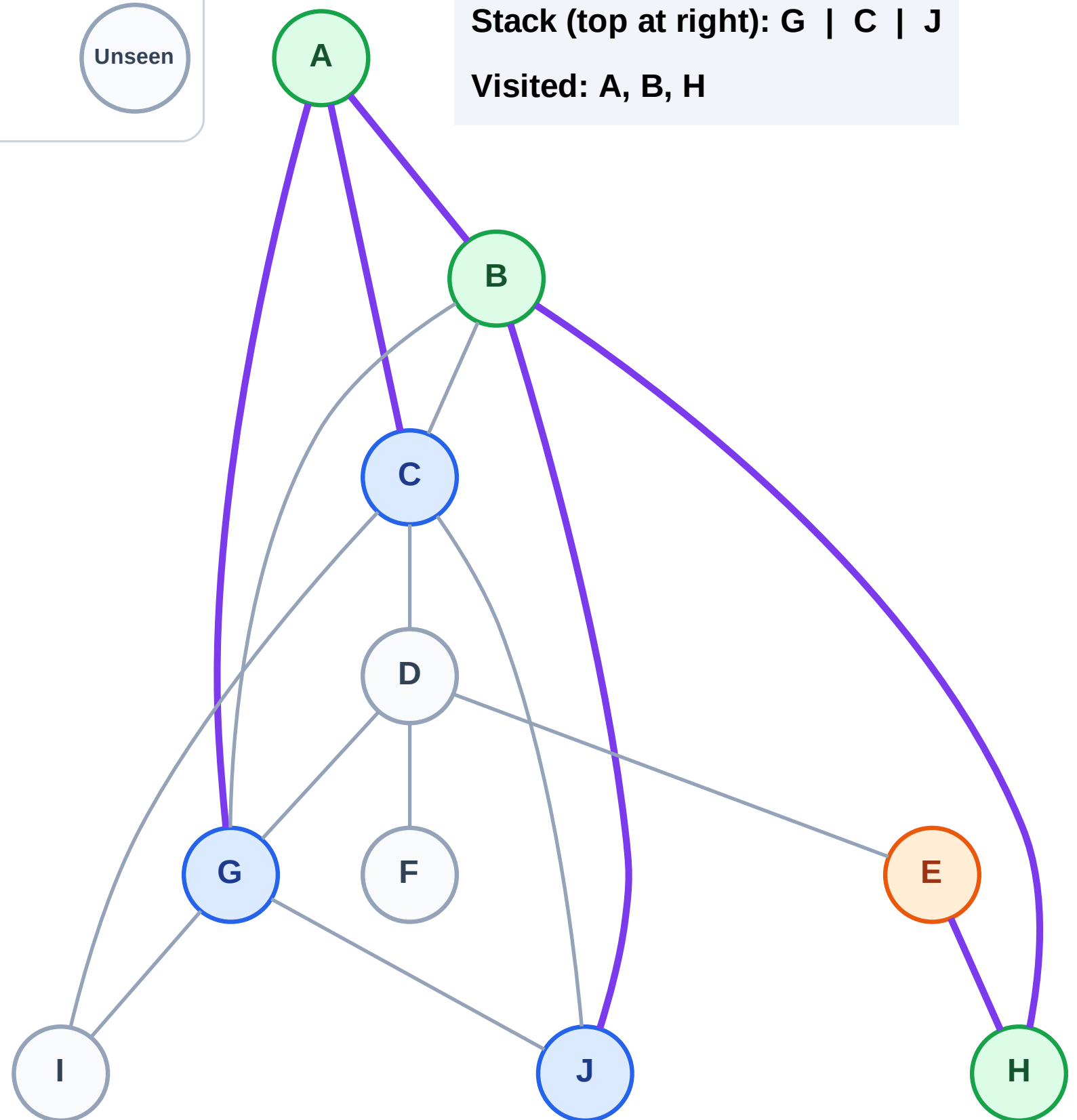
Step 8: Process node E

Legend



Stack (top at right): G | C | J

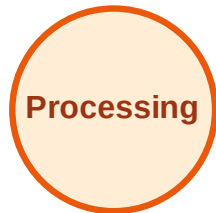
Visited: A, B, H



DFS Traversal

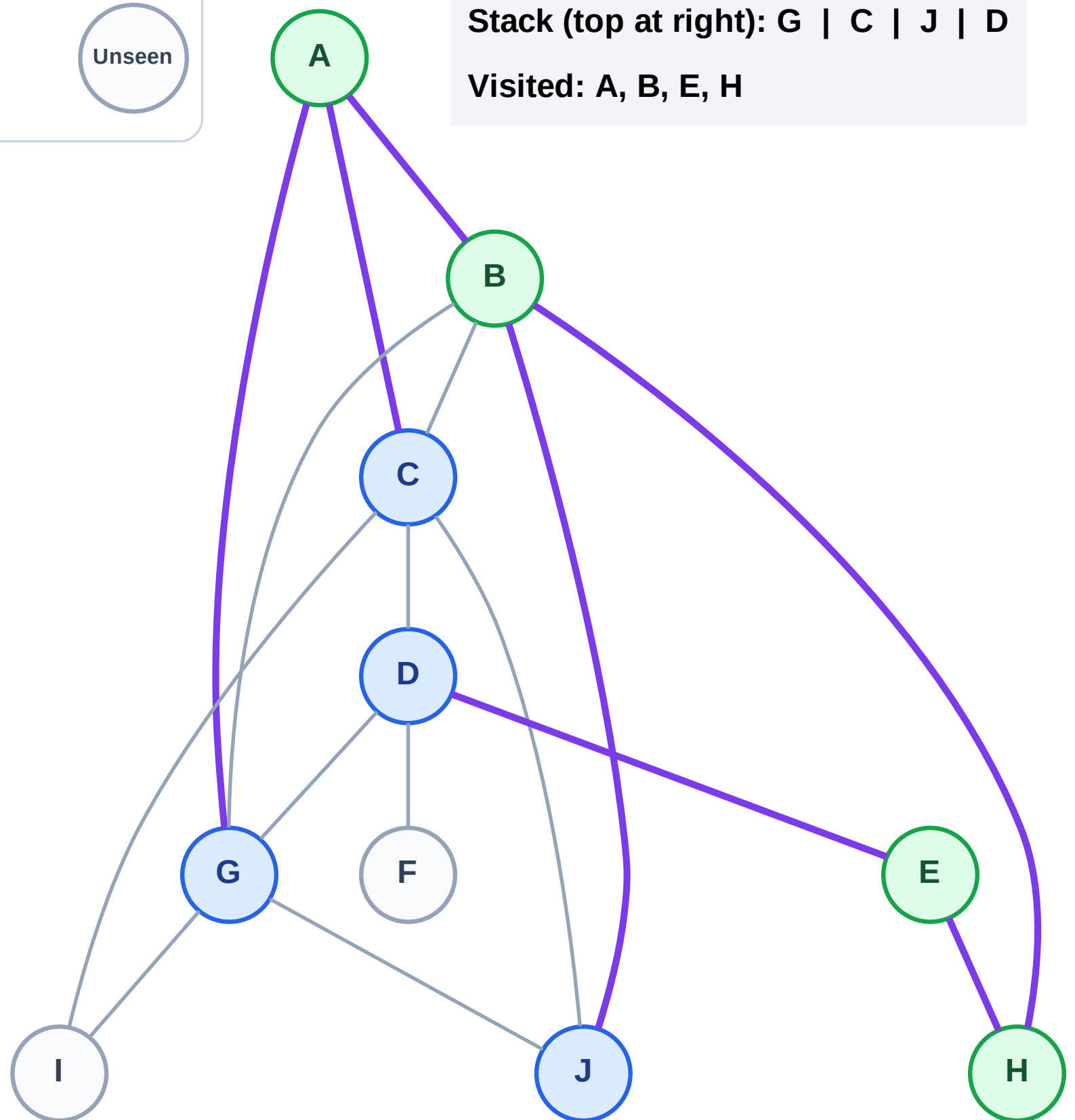
Step 9: Discover D from E

Legend



Stack (top at right): G | C | J | D

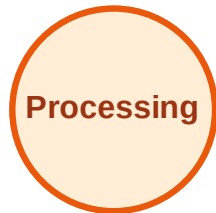
Visited: A, B, E, H



DFS Traversal

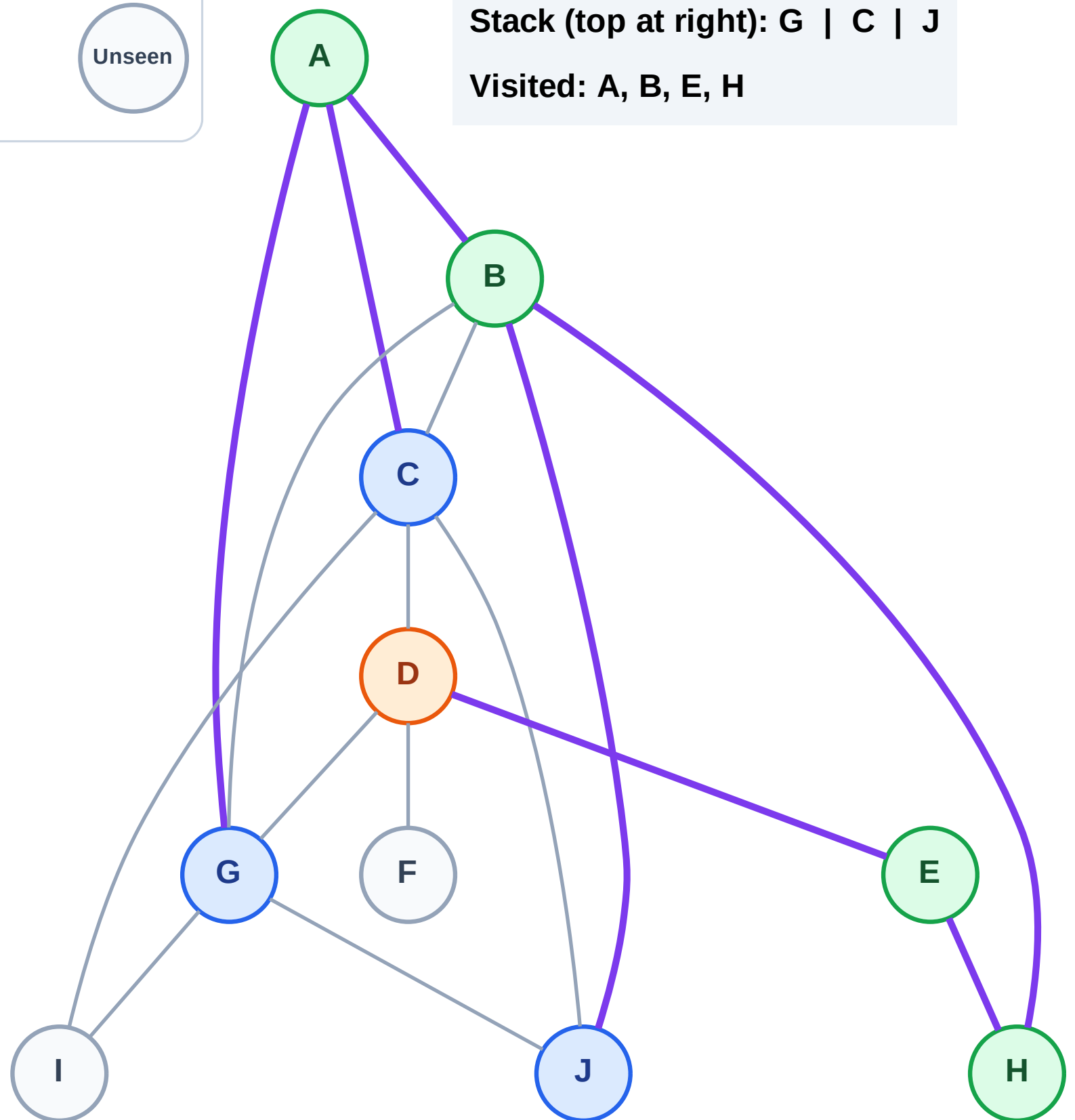
Step 10: Process node D

Legend



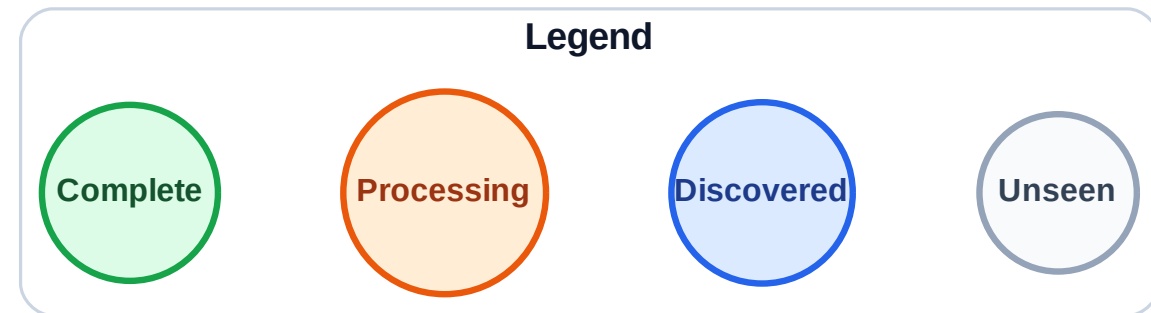
Stack (top at right): G | C | J

Visited: A, B, E, H

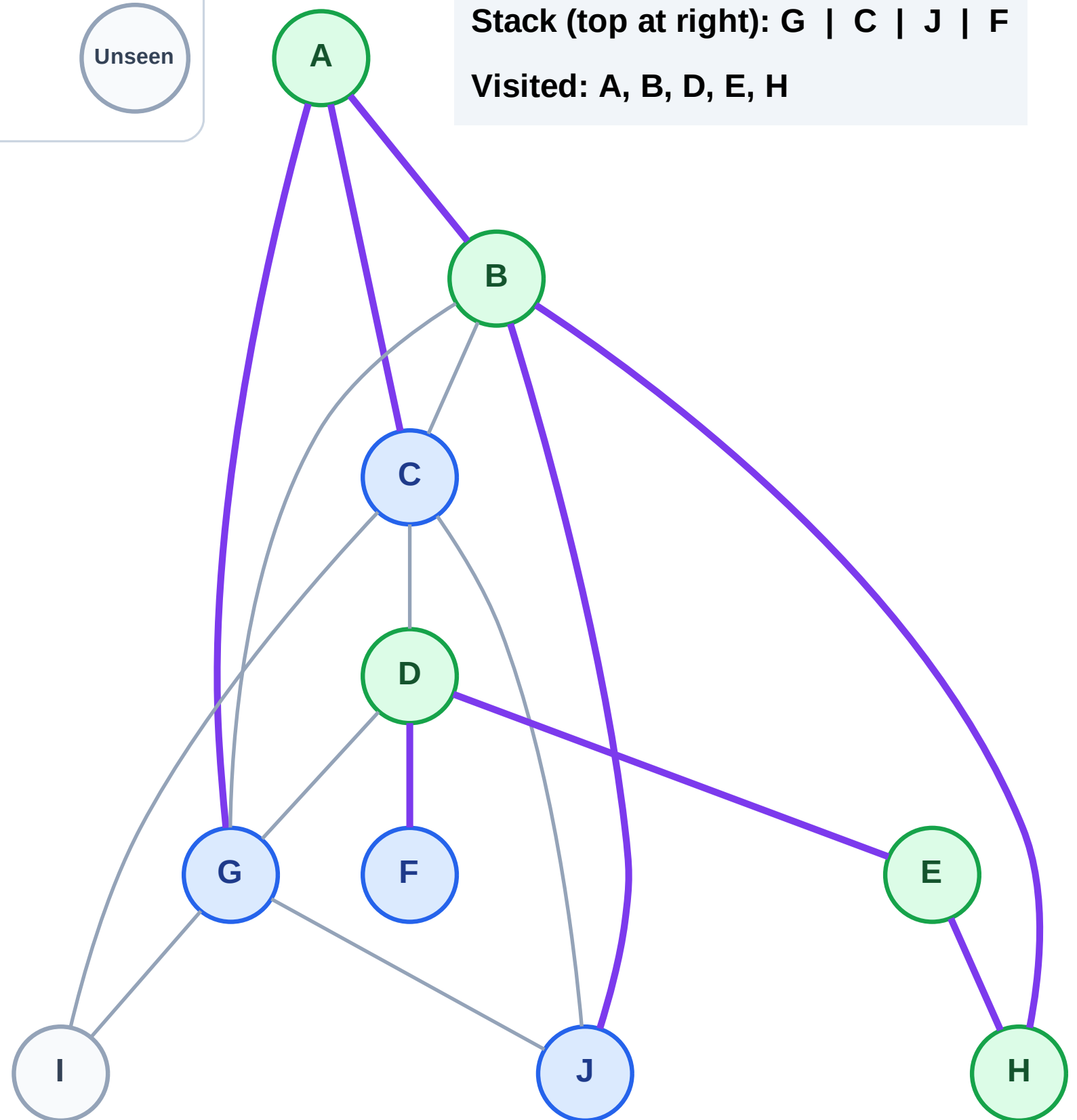


DFS Traversal

Step 11: Discover F from D



Stack (top at right): G | C | J | F
Visited: A, B, D, E, H



DFS Traversal

Step 12: Process node F

Legend

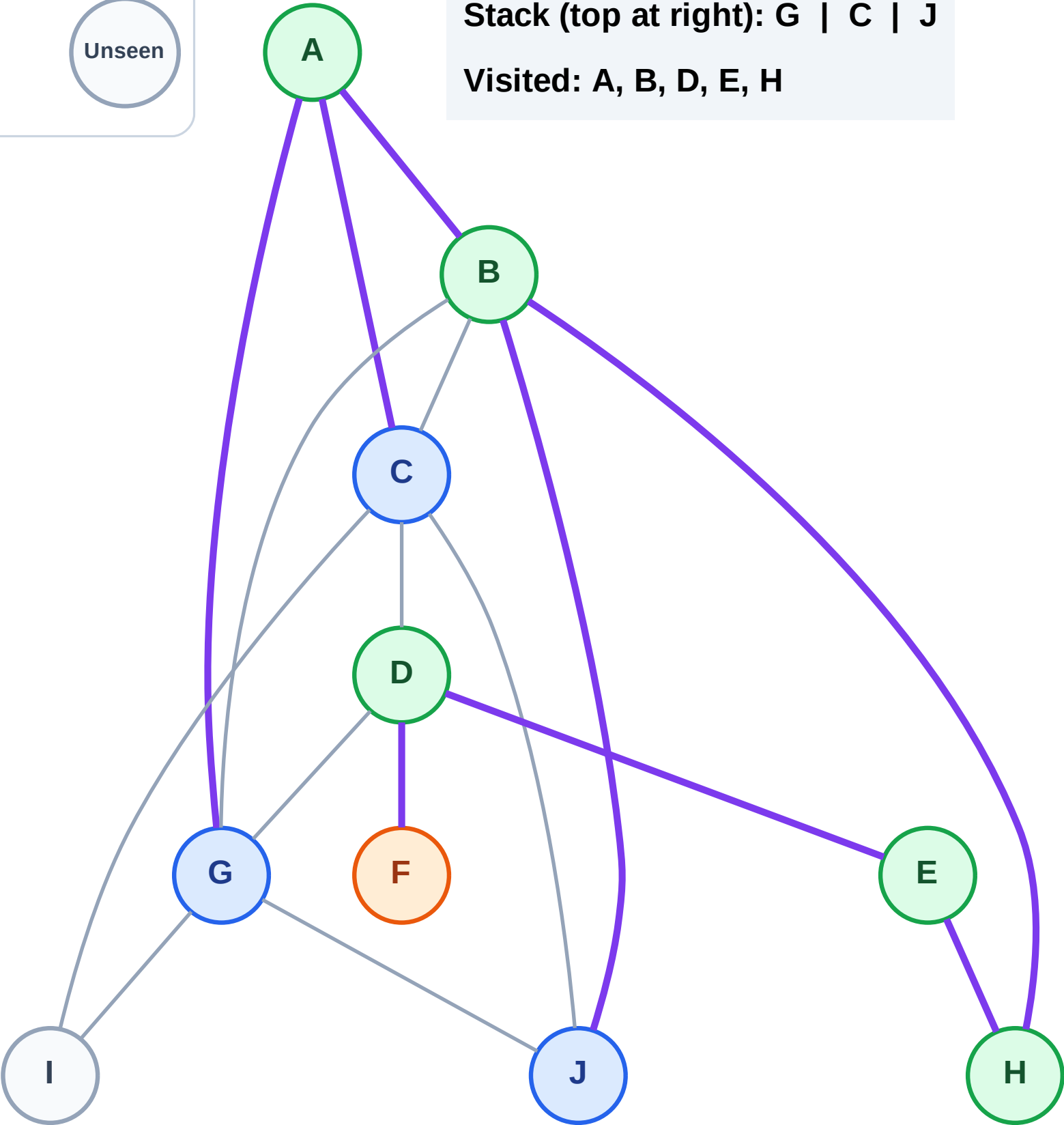
Complete

Processing

Discovered

Unseen

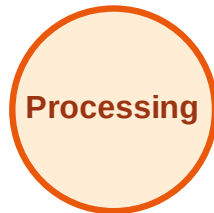
Stack (top at right): G | C | J
Visited: A, B, D, E, H



DFS Traversal

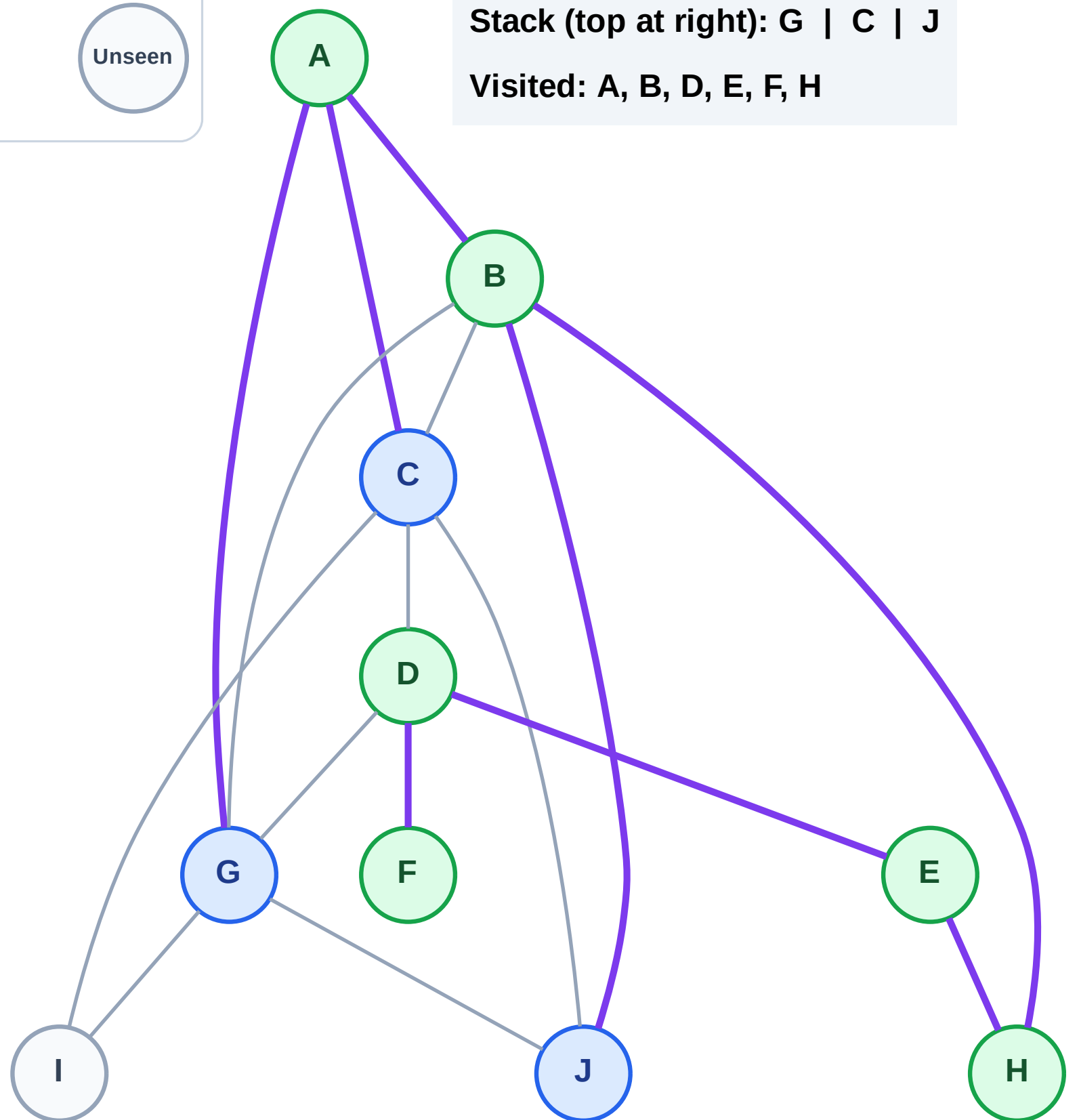
Step 13: No new neighbors from F

Legend



Stack (top at right): G | C | J

Visited: A, B, D, E, F, H



DFS Traversal

Step 14: Process node J

Legend

Complete

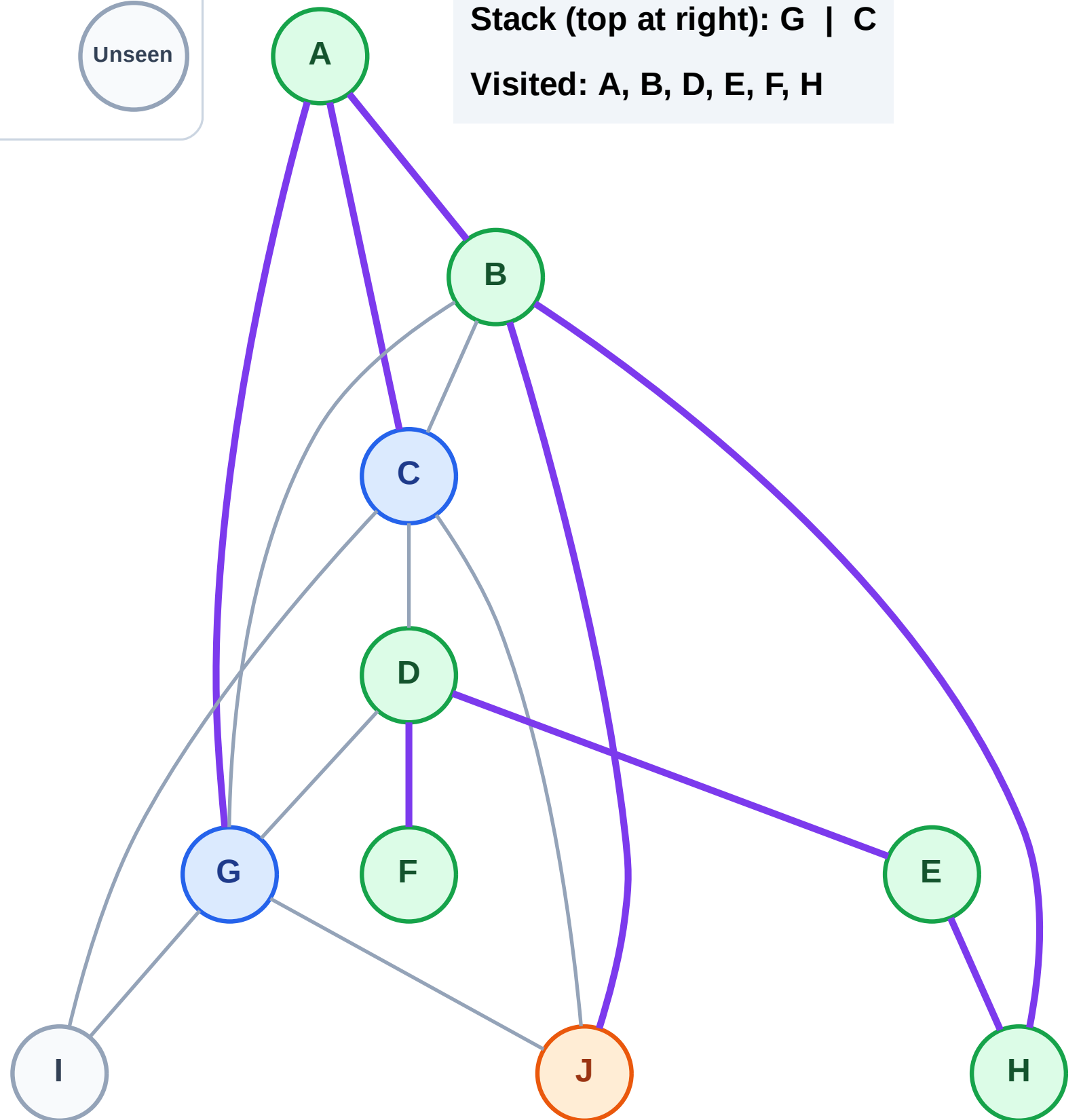
Processing

Discovered

Unseen

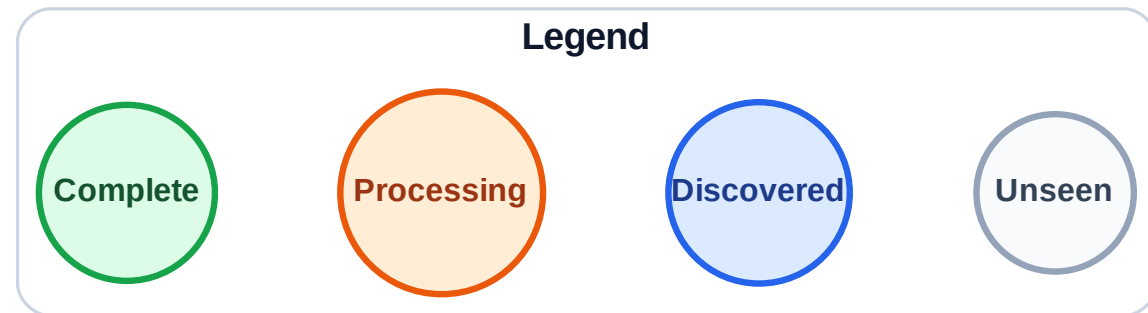
Stack (top at right): G | C

Visited: A, B, D, E, F, H



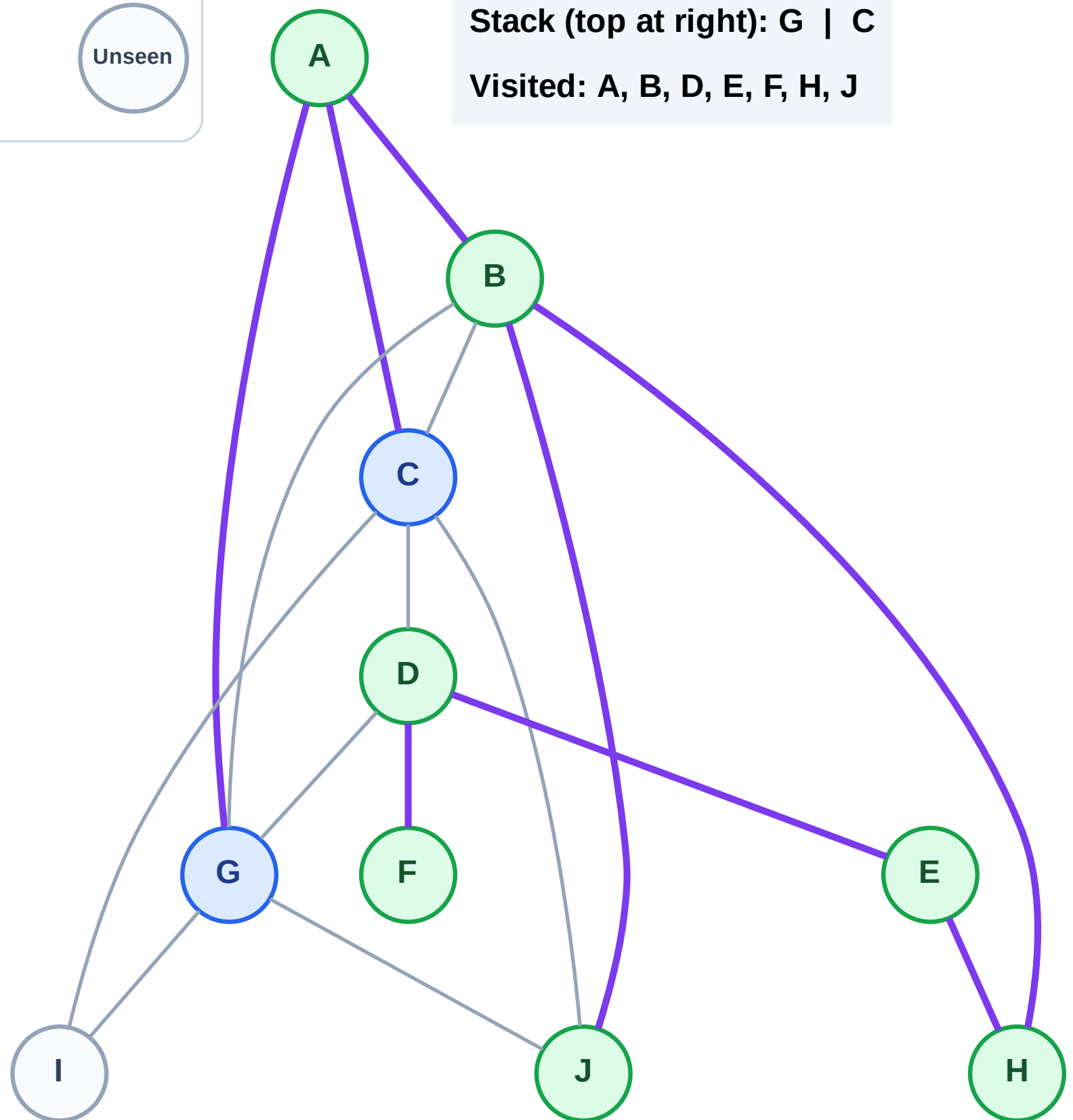
Step 15: No new neighbors from J

Step 15: No new neighbors from J



Stack (top at right): G | C

Visited: A, B, D, E, F, H, J



DFS Traversal

Step 16: Process node C

Legend

Complete

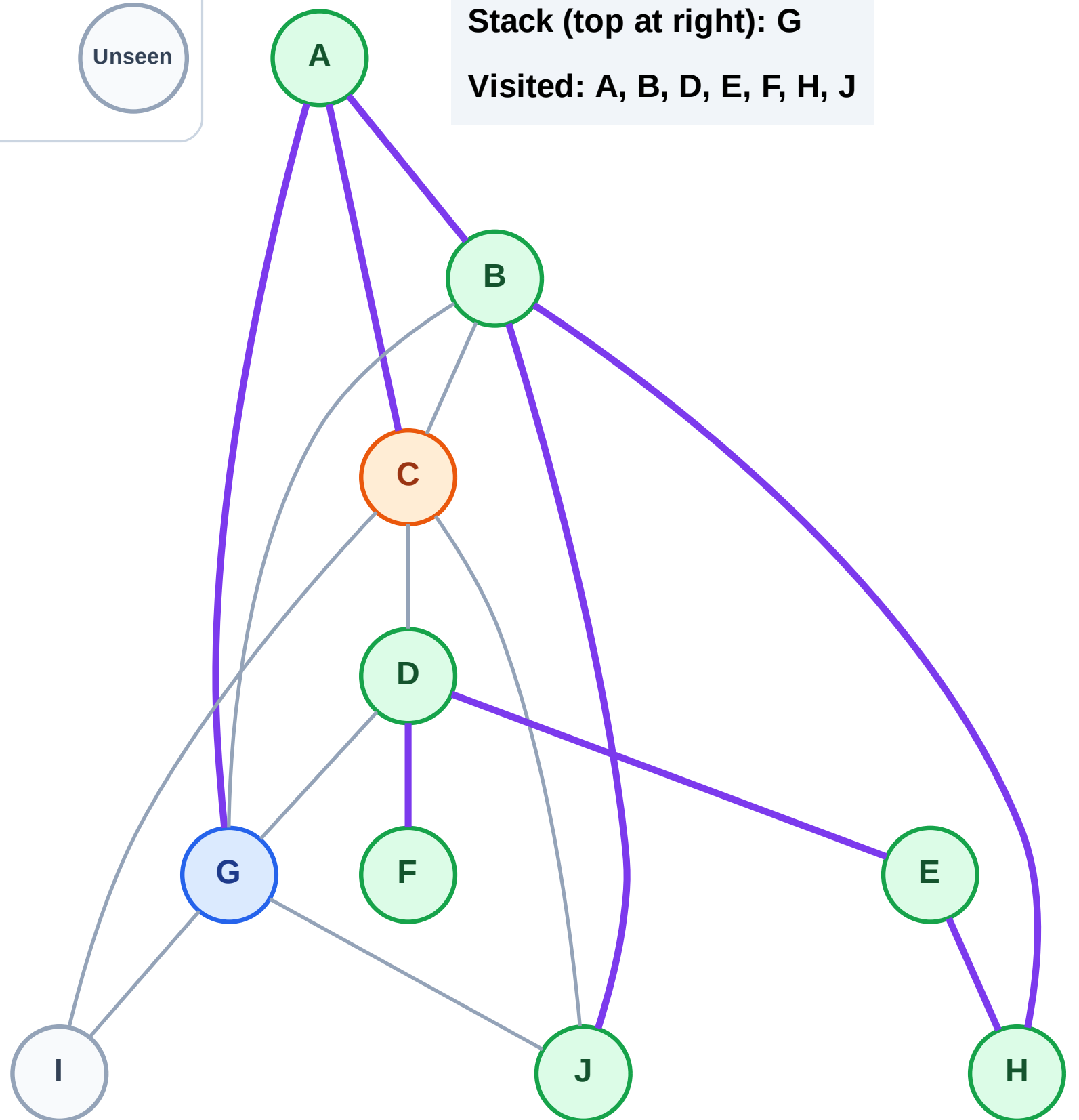
Processing

Discovered

Unseen

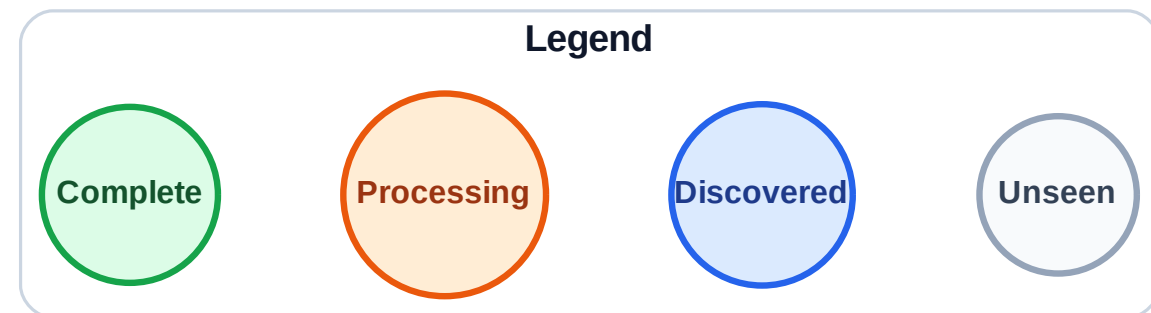
Stack (top at right): G

Visited: A, B, D, E, F, H, J

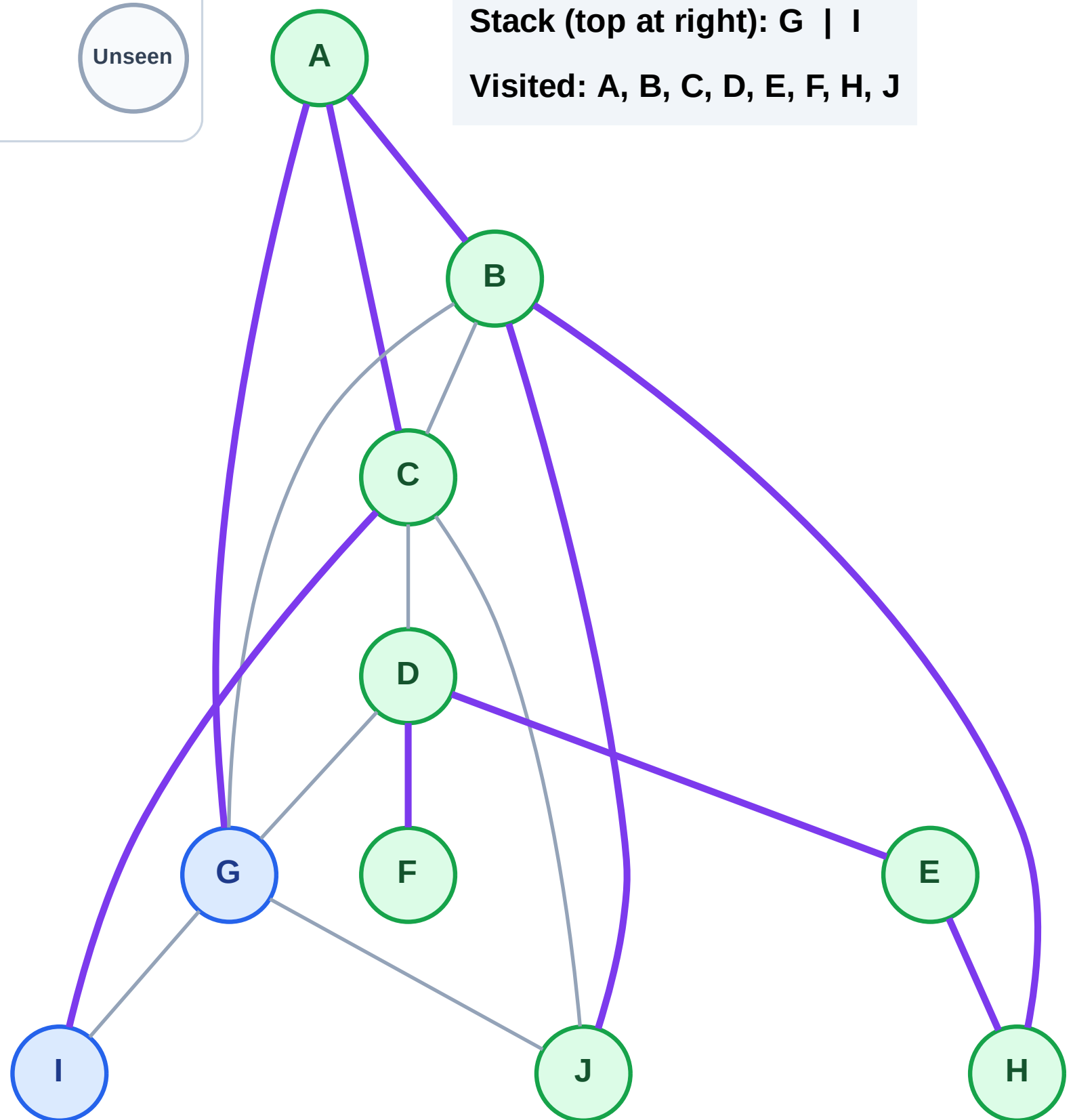


DFS Traversal

Step 17: Discover I from C



Stack (top at right): G | I
Visited: A, B, C, D, E, F, H, J



DFS Traversal

Step 18: Process node I

Legend

Complete

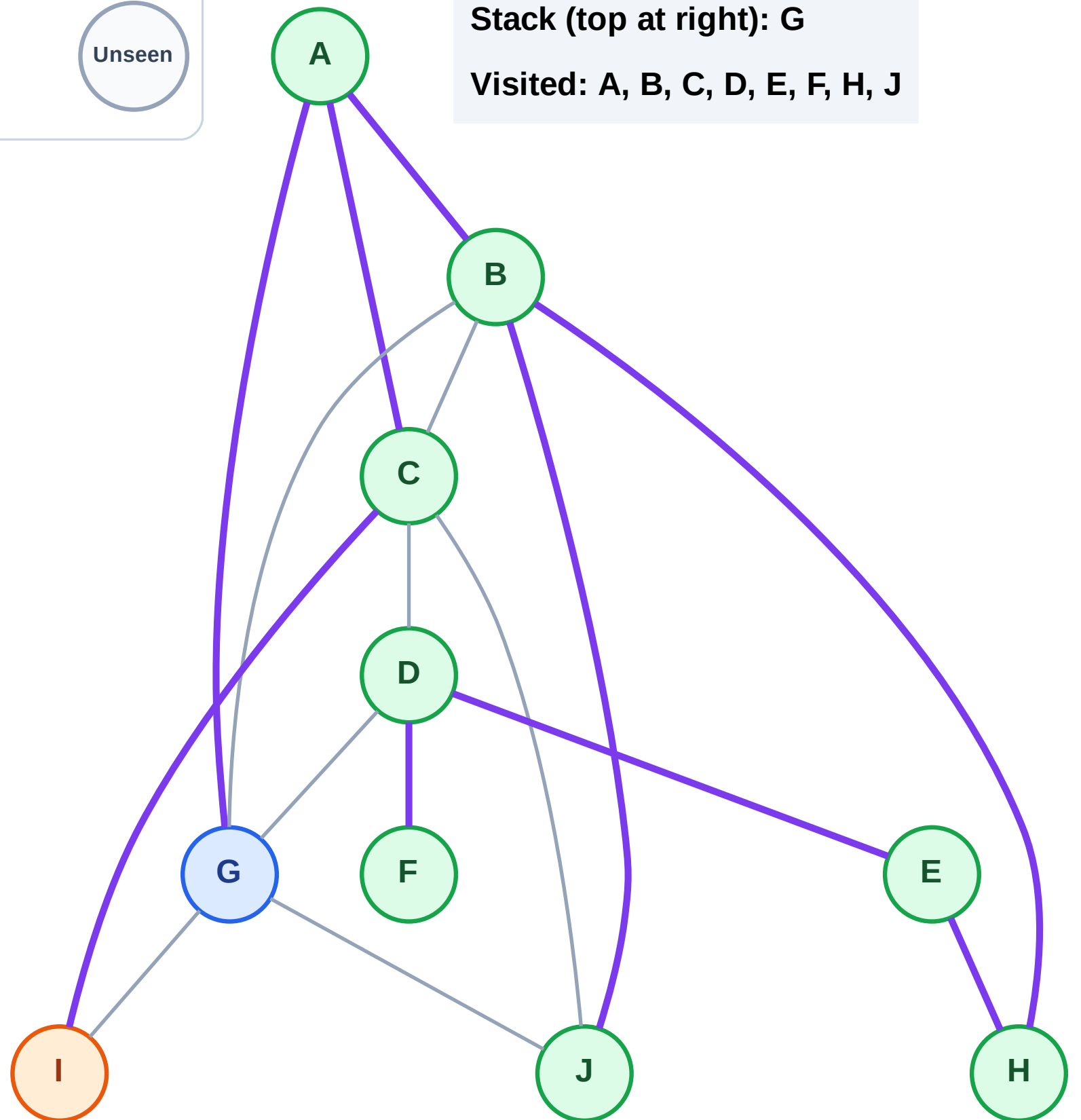
Processing

Discovered

Unseen

Stack (top at right): G

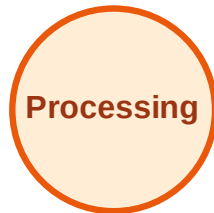
Visited: A, B, C, D, E, F, H, J



DFS Traversal

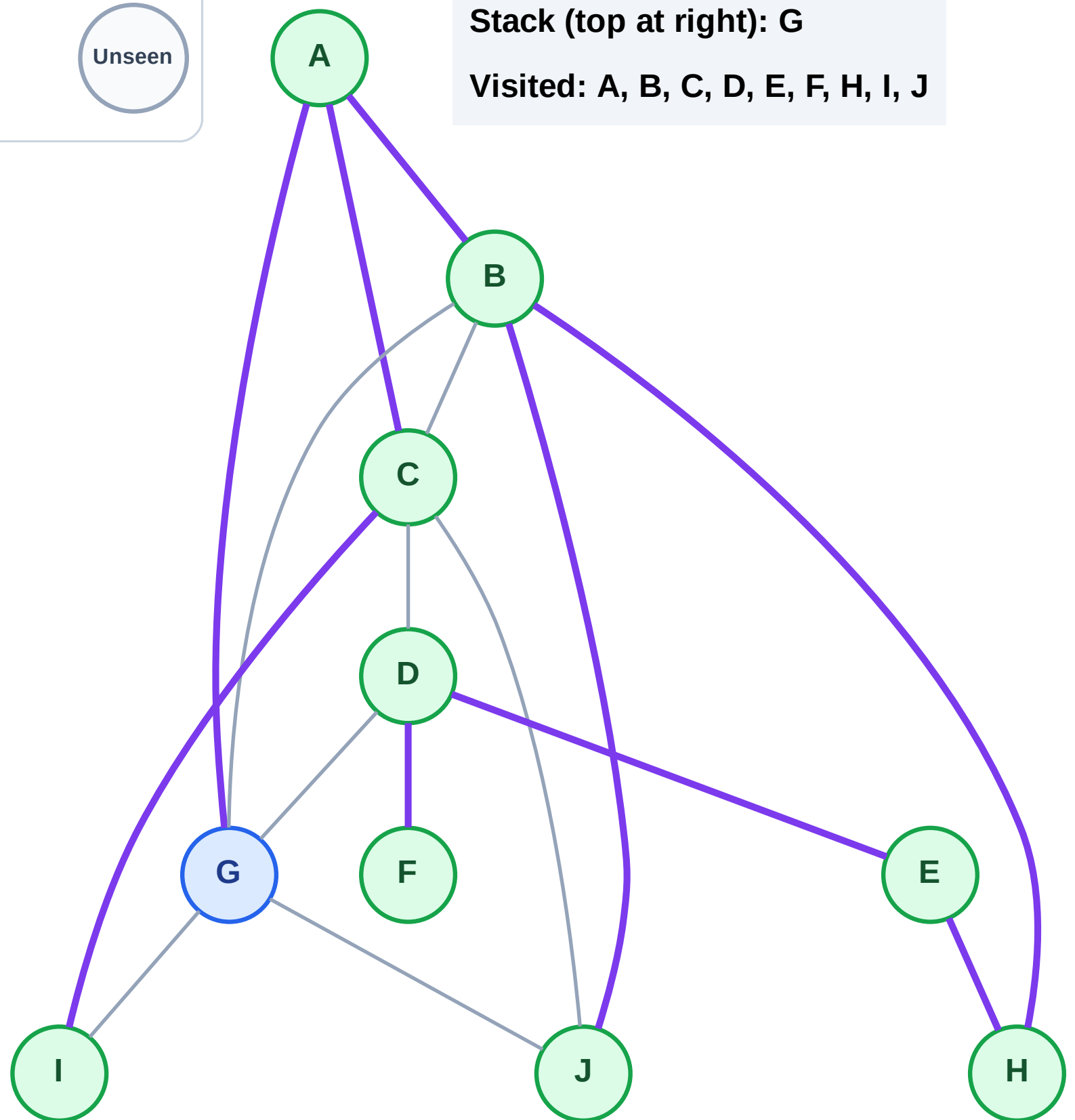
Step 19: No new neighbors from I

Legend



Stack (top at right): G

Visited: A, B, C, D, E, F, H, I, J



DFS Traversal

Step 20: Process node G

Legend

Complete

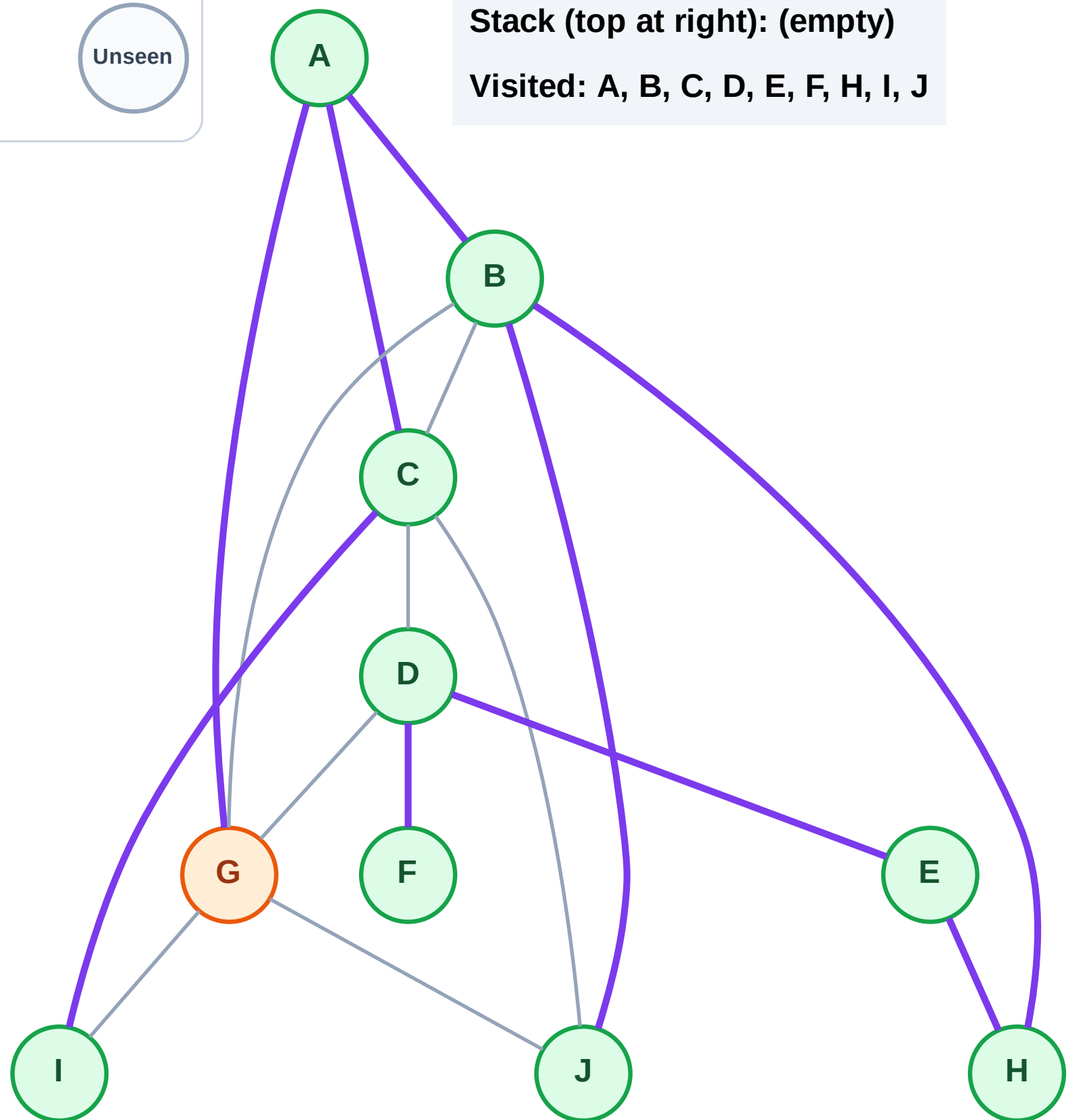
Processing

Discovered

Unseen

Stack (top at right): (empty)

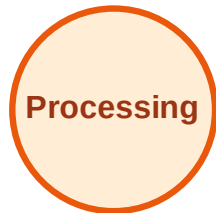
Visited: A, B, C, D, E, F, H, I, J



DFS Traversal

Step 21: No new neighbors from G

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

